

Temporal Stability of Soil Moisture Spatial Pattern and Subsurface Preferential Flow Pathways in the Shale Hills Catchment

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ABSTRACT

Hydropedologic approaches utilize a strategy of “map first, then design” and “direction first, then velocity” in enhancing the understanding of complex landscape processes. This is illustrated in this study by examples dealing with (i) the mapping of soils and landforms in monitoring and interpreting soil moisture dynamics and (ii) the identification of flow pathways in determining landscape water fluxes. Year-round monitoring at 77 sites in the Shale Hills Catchment in central Pennsylvania revealed a temporal stability of soil moisture spatial pattern as governed by soil types and landforms, and suggested the significance of subsurface preferential flow in rapid channeling of precipitation to stream discharge. The five soil series identified in the catchment had the following decreasing trend of moisture storage within the upper 1.1-m solum: Ernest > Blairton ≥ Rushtown ≥ Berks > Weikert. The four landform units showed a decreasing trend of soil moisture storage: Valley > Swale > Hillslope > Hilltop. The 77 monitoring sites exhibited considerable ranking stability throughout the monitoring year at multiple depths, with the subsurface's moisture ranking stability being slightly stronger than that at the surface. A slope-intercept analysis of linear regression further described the four conditions of temporal stability as related to soil moisture and hydrologic dynamics. Because of more extensively distributed deeper soils and hydrologically active swales, plus favorable subsurface lateral flow pathways and slightly higher cumulative rainfall, the south-facing slope in this V-shaped catchment was hydrologically more active than the north-facing slope in terms of draining more water at a faster rate to the stream. Approximately two-thirds of the soil horizons measured in the catchment had lateral saturated hydraulic conductivity (K_{sat}) values 1.5 to 142.5 higher than vertical values. Because of a moderate slope (up to 25–48%), horizontally dipping shale bedrock (11.5–17.1°), and shallow tree rooting systems (branching laterally), subsurface lateral flow was prominent in this humid forested catchment.

IDENTIFICATION AND PREDICTION of soil moisture patterns (spatiotemporal organization) at different scales are important to pedologic and hydrologic studies (Grayson and Blöschl, 2000; Grayson et al., 2002; Lin et al., 2005, 2006a, 2006b). Patterns offer insight regarding the variability of structures and functions over an area, as well as the underlying processes controlling hydrologic response (Grayson et al., 1997, 2002; Lin et al., 2006b). Some spatial patterns of soil moisture are temporally persistent—the notion of “time stability” or “temporal stability” (Vachaud et al., 1985; Kachanoski and de Jong, 1988; Mohanty and Skaggs, 2001; Grant et al., 2004; Pachepsky et al., 2005). Within a field or watershed, soils in some areas may be consistently

wetter or drier than in other areas, or compared with the average moisture content across the whole area. Vachaud et al. (1985) defined temporal stability of spatially measured soil moisture as the time invariant association between spatial location and classical statistical parameters of a soil property (such as spatial average or relative ranking). Kachanoski and de Jong (1988) expanded the definition to include a general linear transformation in time and spatial-scale-dependent time stability. They pointed out that the concept of time stability is a reflection of the temporal persistence of spatial structure over a sampling domain (which is scale dependent), and that temporal stability is different from temporal autocorrelation and autoregression equations (TAAEs) for temporal stochastic modeling. While TAAEs hold for all equal time periods that reflect the magnitude of temporal change, time stability does not deal with the similarity of temporal changes. Rather, it indicates that once a change has occurred it will be distributed across the spatial domain in a specific pattern (Kachanoski and de Jong, 1988). This concept of temporal stability enhances the understanding of the interaction between spatial and temporal scales of soil hydrologic properties and their relationships to soil and landscape features. This concept also offers the potential of reducing a large measurement network to fewer representative locations. For example, Vachaud et al. (1985) showed that some locations over three European agricultural fields conserved the property to represent the mean, standard deviation, and extreme values of the field volumetric water content (θ) at any time of the year, hence potentially reducing the number of measurements needed to characterize the normal probability distribution of θ . Grant et al. (2004) also indicated that estimates of catchment mean and standard deviation of soil water storage to a depth of 0.75 m may be characterized by relatively few measurements because of significant temporal stability observed across the 0.36-km² mountain catchment they studied.

Temporal stability of moisture content at the same soil depth has been studied by various researchers (e.g., Vachaud et al., 1985; Kachanoski and de Jong, 1988; Zhang and Berndtsson, 1991; Goovaerts and Chiang, 1993; Ferreyra et al., 2002). As Martinez-Fernandez and Ceballos (2003) and Pachepsky et al. (2005) pointed out, relatively less is known about the temporal stability of soil moisture as a function of depth. A priori one would expect greater stability at the bottom of a soil profile because of reduced dependence on climatic and biological factors. Cassel et al. (2000) attributed the greater

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Abbreviations: DEM, digital elevation model; TAAE, temporal autocorrelation and autoregression; TDR, time domain reflectometry; 3M, mapping, monitoring, and modeling.

temporal stability in deeper soils in an agricultural field to reduced crop root water uptake. Pachepsky et al. (2005) also reported the weakest time stability at the shallowest observation depth among the five depths measured (0.15–0.95 m). Martinez-Fernandez and Ceballos (2003), on the other hand, did not observe specific patterns of stability with respect to soil depth. They did, however, observe that temporal stability was always higher when the soils were dry at all depths that they studied (0.05, 0.25, 0.50, and 1 m) and that the lowest temporal stability coincided with transition from dry to wet conditions. However, Zhang and Berndtsson (1988) and Hupet and Vanclooster (2002) documented weaker temporal stability during dry periods compared with wet periods.

A common belief regarding the soil moisture spatial pattern of a landscape is that topography becomes increasingly important in wet periods, but during dry periods soil moisture patterns depend primarily on soil properties, with topography having a limited effect (e.g., Grayson and Blöschl, 2000). However, other studies (e.g., Chappell et al., 1990; Crayosky et al., 1999; Sidle et al., 2000) showed that riparian zones and geomorphic hollows, having higher soil moisture during dry periods compared with other watershed positions, are dominant suppliers of storm runoff even during dry periods. Nevertheless, the controlling factors leading to the temporal stability of soil moisture spatial pattern and its depth function are not yet well understood. Furthermore, few studies have addressed soil moisture spatiotemporal patterns in areas larger than a few hectares and for time periods longer than a few months (Grayson and Western, 1998; Martinez-Fernandez and Ceballos, 2003). It is therefore the purpose of this study to investigate the spatial distribution and depth function of soil moisture temporal stability and their controlling factors throughout a 7.9-ha forested catchment for a 1-yr period.

Mapping soils and landforms is important for capturing and explaining the temporal stability of soil moisture spatial pattern and can provide better insights into the space–time dynamics of flow systems than point data. Because we can rarely sample densely enough to fully capture the spatial–temporal variability of a watershed, we need to identify patterns that link point observations to areal phenomena and use such knowledge to optimally design sampling and modeling. Key patterns for understanding soil moisture dynamics and landscape hydrologic processes include spatial distributions of soil types and landforms as well as their interrelationships. We normally monitor pedons to collect point data and model landscapes trying to understand areal distributions. A key feature connecting the two is the mapping of the distribution of various soils over the landscape. Relatively static properties of soil and landscape features (such as topography and soil type) may be mapped to assist in scaling and modeling of landscape–soil–water relationships, while more dynamic properties (such as soil moisture and stream flow) can be monitored to calibrate and refine model predictions. Chappell and Ternan (1992) demonstrated the importance of soil spatial structure in hydrologic modeling. They pointed

out that the evidence for soil profile and catena structure within soil hydrologic properties and resultant solute flow paths has not been fully embraced by the hydrologic community.

The Landscape Processes symposium held at the 2004 annual meeting of the Soil Science Society of America raised the awareness of the landscape perspective in solving many environmental and agricultural problems. The symposium also highlighted the necessity of connecting “mapping, monitoring, and modeling” (3M) in understanding landscape processes. The iterative loop of this “3M” cycle has been suggested as a useful hydro-pedologic approach to landscape studies (Lin et al., 2005). The inherent strategy includes: (i) map first, then design and (ii) direction first, then velocity. The first suggests that detailed mapping of soils and landforms can provide a meaningful stratification of the landscape for optimal design for sampling and monitoring, and that adequate attention needs to be paid to soil morphology and horizonation because they reveal soil structure and flow heterogeneity. The second strategic aspect emphasizes the importance of getting flow pathways right when measuring and modeling fluxes, something that many existing hydrologic models fail to do. For example, many current hydrologic models poorly predict subsurface lateral flow, and few studies have investigated soil lateral conductivity (Wood, 1999; Leibundgut et al., 1999; Lin et al., 2006b). However, sloping topography, stratigraphy, and soil layering (especially water-restricting layers) all favor lateral flow (Richardson et al., 2001; Uchida et al., 2003; Lin et al., 2005a). In fact, experiments of Hewlett and Hibbert (1963) and Whipkey (1965) enlightened many hydrologists to the importance of subsurface lateral flow in hillslope hydrology. However, this process has not yet been well incorporated into many existing hydrologic models (Leibundgut et al., 1999). Describing lateral flow adequately is an important step in quantifying key runoff mechanisms and in improving subsurface flow models. For example, Brooks et al. (2004) demonstrated that hillslope-scale lateral saturated hydraulic conductivity (K_{sat}) was between one and two orders of magnitude larger than small-scale vertical K_{sat} measured with constant and falling head permeameters and Guelph permeameter. As interests shift to issues involving the transport of solutes and sediments driven by water flow, hydrologic models need to predict first and foremost the flow pathways correctly and then the associated fluxes (the notion of “direction first, then velocity”); that is, they must be accurate for the right reasons, something that is not necessarily needed for acceptable predictions of integrated catchment runoff at the watershed scale (Grayson and Blöschl, 2000; Sidle, 2004; Lin et al., 2006b). This also has strong implications for assessing distributed management practices in watersheds.

The objective of this paper is to report some initial findings from a hydro-pedologic observatory established for investigating fundamental processes of landscape water fluxes at multiple scales and for characterizing spatiotemporal patterns of soil moisture and their relations to soil and landscape features. Specifically, this

paper will (i) examine the temporal stability of soil moisture spatial pattern as governed by soil types and landforms in a forested catchment and (ii) investigate landscape water flow pathways in relation to soil characteristics and landscape features. Year-round soil moisture data collected at 77 monitoring sites in a forested catchment were analyzed, along with stream discharge, precipitation, in situ observations, and laboratory characterizations of soil lateral and vertical hydraulic conductivities.

MATERIALS AND METHODS

The Study Catchment and Mapping of Its Landforms and Soils

The 7.9-ha forested Shale Hills Catchment in central Pennsylvania (Fig. 1) is characteristic of low-lying shale hills of the Ridge and Valley Physiographic Province of the eastern United States that extends from central New York to northern Alabama. This catchment is V-shaped and characterized by moderate slopes (up to 25–48%) and narrow ridges. The catchment is covered by deciduous trees (mostly maple [*Acer*

spp.], oak [*Quercus* spp.], and hickory [*Carya* spp.]) on the sloping areas and ridges, and by eastern hemlock conifers [*Tsuga canadensis* (L.) Carrière] on the valley floor. The entire catchment is underlain by the thick (>200 m) Rose Hill shale (Berg et al., 1980). Based on our field measurements, depth to bedrock ranges from <0.25 m on the ridge tops and upper side slopes to >2 m in the valley bottom and swales. A depth of bedrock map was developed in this study based on 223 observations recorded during soil survey and instrument installation and interpolated using a third-order local polynomial using ArcGIS Geostatistical Analyst (Lin et al., 2006a).

There are four basic landforms in this catchment (Fig. 1 and 2): (i) south-facing slope with deciduous forest and underbrush, (ii) north-facing slope with deciduous forest and thicker underbrush, (iii) valley floor or floodplain of a first-order headwater stream (a tributary of the Shavers Creek that reaches the Juniata River and onto the Susquehanna River) with evergreen trees along the western side and deciduous forest on the eastern side, and (iv) topographic depressional areas (swales) with deciduous forest cover and deeper soils. The valley is oriented in an east–west direction separating north-facing and south-facing slopes (Fig. 1). Elevation of the area ranges from 256 m at the outlet of the catchment to 310 m at the highest ridge. The relatively uniform side slopes are

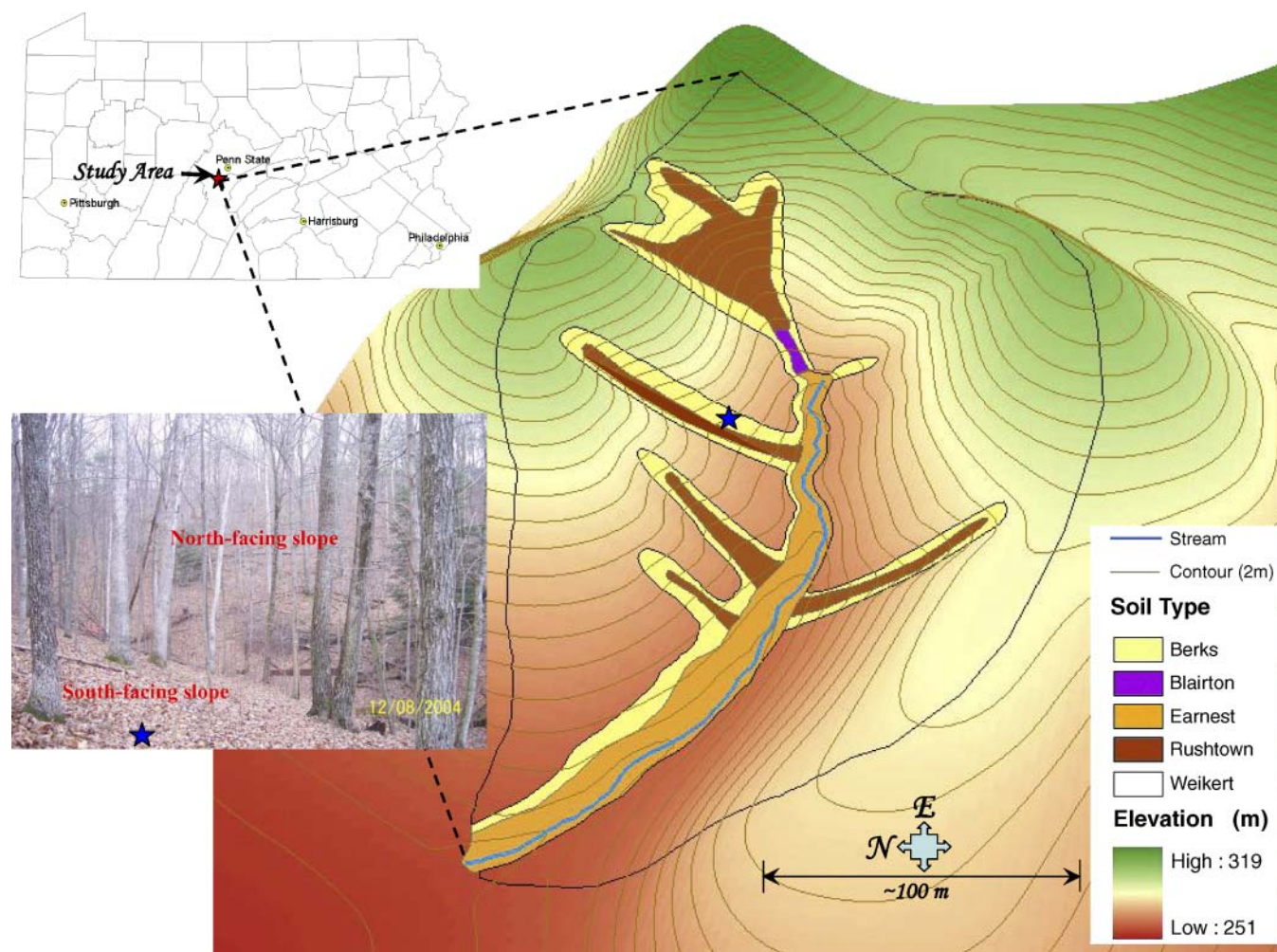


Fig. 1. The V-shaped Shale Hills Catchment (rendered three-dimensionally) located in central Pennsylvania. The soil map depicts the five soil series identified in the catchment, with four soil series colored differently and the Weikert series covering the rest of the catchment (not colored in order to show the elevation variation). The inset photo shows a landscape view downslope from a point indicated by the blue star. Note the scale bar indicated is only an approximation because the three-dimensional rendering distorted the actual scale.

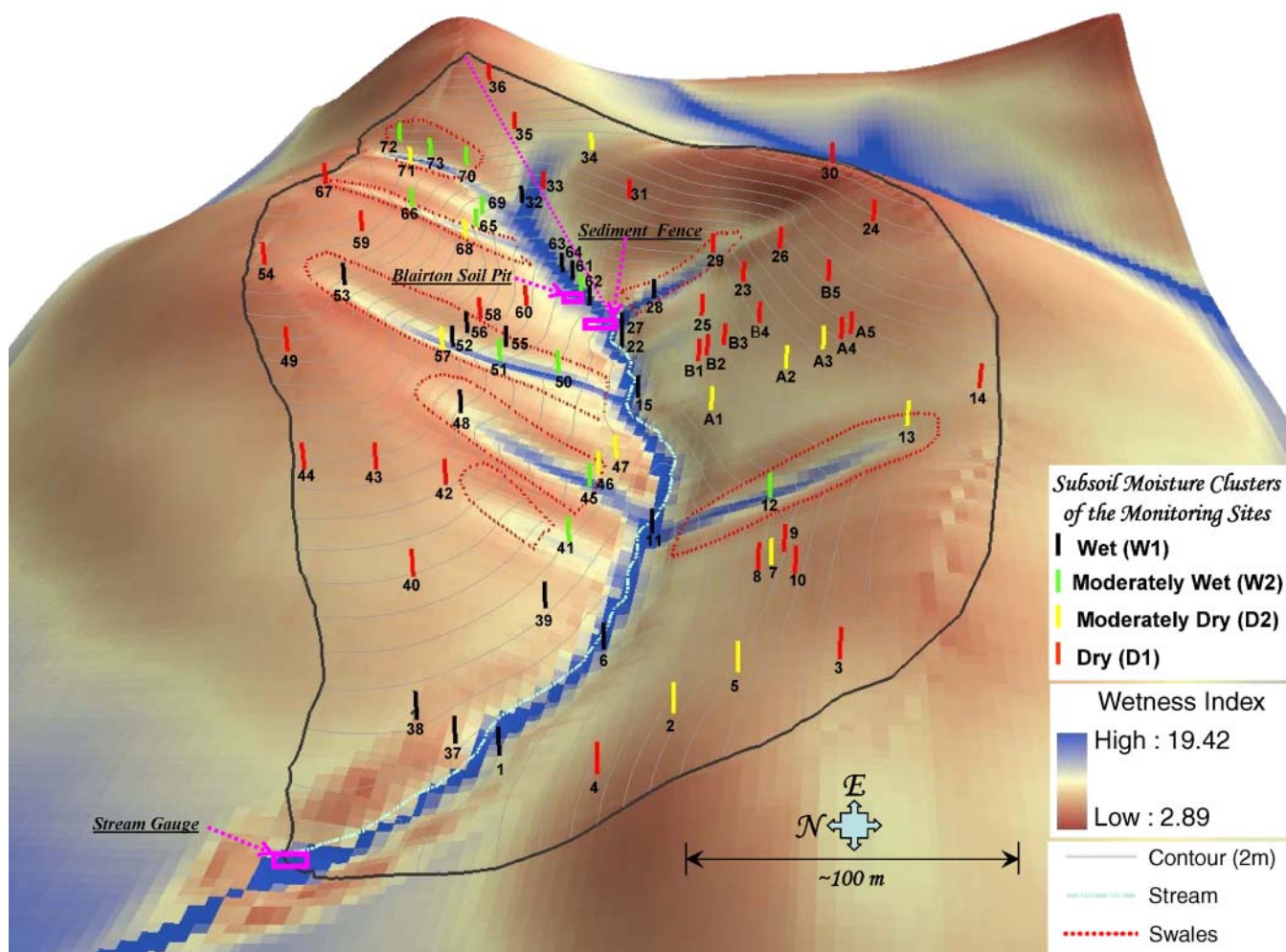


Fig. 2. The 77 monitoring sites established in this study. The numbers correspond to the site numbers listed in Table 2. The subsoil wetness clusters (wet, moderately wet, moderately dry, and dry) were based on a combined consideration of soil thickness (depth to bedrock), topographic wetness index, and local slope. The background map is the topographic wetness index calculated with Eq. [2]. The red dashed polygons are topographic depressions (swales). Note the scale bar indicated is only an approximation because the three-dimensional rendering distorted the actual scale.

periodically interrupted by seven distinct swales of varying sizes on both sides of the stream; five are on the south-facing side and two are on the north-facing side (Fig. 2). The existing 10-m resolution digital elevation model (DEM) for the catchment originally did not show the swales well. Consequently, we conducted a local survey of swales and ridges to refine the elevation model using a Trimble (Sunnyvale, CA) Pro XR GPS receiver and postprocessed using a base station to achieve optimal accuracy. This field survey and data processing included (i) GPS of swale boundaries and slope breakpoints along the center of the swales, (ii) recording angle between swale center points and edges using a field clinometer, (iii) calculating the planimetric distance between the swale center points and edge points and vertical displacement from edge to center of the swales, (iv) determining the function that describes the elevation profile of each swale, (v) resampling the 10-m DEM to 3 m, (vi) running a 3×3 low pass filter to smooth the surface, and finally (vii) using the AGREE method (Maidment, 2002) of DEM conditioning to “burn” the stream vector into the elevation surface.

The soils of the catchment were formed from shale colluvium or residuum, with many channery shale fragments throughout most of the soil profiles (Lin et al., 2006a). Soils on the hillsides have a shallow depth to bedrock, while the valley floor and swale areas have deeper depth to the shale bedrock.

Soils on the hillslopes generally have silt loam texture, moderately developed soil structure, high permeability, and are well drained (Table 1). Redoximorphic features were only found in soils along the valley floor and occasionally at the lower portion of some swales as a result of seasonal soil saturation. Since the entire catchment is covered by forest, all soils have an approximately 0.05-m-thick organic layer (Oe horizon) comprised of decaying leaf litters and other organic materials. Such an organic layer acts like a sponge and has prevented overland flow from occurring in this steep terrain (Lin et al., 2006a).

A detailed soil map was developed for the catchment using standard grid-transect method (Soil Survey Division Staff, 1993), supplemented by additional augering and ground penetrating radar investigations to refine the soil boundaries (particularly in the swale areas). A total of five soil series (Weikert, Berks, Rushtown, Blairton, and Ernest series) were identified and mapped (Fig. 1). The Rushtown, Blairton, and Ernest series were the closest established official soil series that matched reasonably (but not completely) with the soils identified at Shale Hills. For the lack of new soil series names, we adopted these existing series names in this study. Depth to bedrock and landscape positions were the main criteria used to differentiate these five soil series. The Weikert series (loamy-

Table 1. Basic characteristics of the five soil series identified at the Shale Hills Catchment.

Horizon	Depth m	Texture	Rock fragments (shale channery)	Pedality [†] (soil structure)	Special features	Bulk density Mg m ⁻³	Total porosity %	Saturated hydraulic conductivity (<i>K</i> _{sat})		<i>K</i> _{sat} ratio	
								Vertical	Horizontal	Horizontal over Vertical	Vertical A horizon over 1st B horizon
<u>Weikert Series</u>											
Oe	0-0.05										
A	0.05-0.12	silt loam	0	2 f gr		0.425	0.840	1.863	ND [‡]	ND	
Bw	0.12-0.24	silt loam	60	2 f sbk		0.960	0.638	2.280	ND	ND	
CR	0.24-0.37	silt loam	90	1 f sbk		1.260	0.525	0.409	0.701	1.72	5.58
R	0.37+										
<u>Berks Series</u>											
Oe	0-0.05										
A	0.05-0.08	silt loam	0	2 f gr		0.455	0.828	13.589	ND	ND	
Bw1	0.08-0.14	silt loam	2	1 vf sbk		1.302	0.509	1.905	0.382	0.20	
Bw2	0.14-0.53	silt loam	2	1 f abk		1.265	0.523	2.027	0.131	0.06	0.94
Bw3	0.53-0.69	silty clay loam	50	1 f abk		ND	ND	0.038	0.018	0.48	
C	0.69-1.45	silty clay loam	90	massive		1.670	0.370	0.062	0.186	3.00	
R	1.45+							0.042	0.020	0.48	
<u>Rushtown Series</u>											
Oe	0-0.05										
A	0.05-0.11	silt loam	5	3 f gr		0.425	0.840	6.374	ND	ND	
Bw1	0.11-0.17	silt loam	5	2 f sbk		ND	ND	0.432	0.684	1.58	
Bw2	0.17-0.26	silt loam	5	1 m sbk		1.204	0.545	0.334	0.493	1.47	
Bw3	0.26-0.38	silty clay loam	5	1 m sbk		1.292	0.512	0.528	0.262	0.50	1.29
BC	0.38-0.60	silty clay loam	50	massive		1.537	0.420	0.979	0.170	0.17	
C	0.60-1.78+	silty clay loam	80	massive		1.637	0.382	0.087	0.259	2.97	
						1.652	0.376	0.080	4.520	56.24	
<u>Blairton Series</u>											
Oe	0-0.05										
A	0.05-0.11	silt loam	<2	3 m sbk parting to 2 f gr		0.550	0.792	0.415	4.637	11.17	
BA	0.11-0.18	loam	<2	2 f sbk		1.180	0.555	0.345	1.704	4.93	
Bt1	0.18-0.29	clay loam	<2	1 m sbk	<5% distinct patchy clay films	1.299	0.510	0.020	0.128	6.42	17.29
Bt2	0.29-0.80	clay loam	10 (very soft)	1 c sbk	<5% distinct patchy clay films	1.372	0.482	0.002	0.284	142.52	
CB1	0.80-1.10	sandy clay loam	80 (very soft)	1 c sbk	2% fine distinct Fe-depletions and Fe-concretions	1.662	0.373	0.018	0.649	36.75	
CB2	1.10-1.52+	sandy clay loam	80 (very soft)	1 c sbk	2% faint discontinuous clay films	1.730	0.347	0.121	0.191	1.58	
					5% fine prominent Fe-depletions and Fe-concretions						
					2% faint discontinuous clay films	1.739	0.344	0.007	0.002	0.27	
<u>Ernest Series</u>											
Oe	0-0.05										
A	0.05-0.15	silt loam	0	1 c sbk parting to 2 f gr		0.856	0.677	0.237	0.459	1.93	
AE	0.15-0.20	silt loam	0	1 f gr		0.865	0.674	0.678	1.338	1.97	
Bw	0.20-0.30	silty clay loam	0	1 m sbk		1.107	0.582	0.271	0.403	1.49	2.50

Continued next page.

Table 1. Continued.

Horizon	Depth m	Texture	Rock fragments (shale channery)	Pedality [†] (soil structure)	Special features	Bulk density Mg m ⁻³	Total porosity	Saturated hydraulic conductivity (K_{sat})		K_{sat} ratio	
								Vertical	Horizontal	Horizontal over Vertical	Vertical A horizon over 1st B horizon
Bt	0.30–0.53	silty clay	0	1 c pr parting to 1 c sbk	20% medium prominent redox and concentrations 60% distinct continuous clay films	1.267	0.522	0.367	0.820	2.23	
2C	0.53–0.89	sandy loam	80 (soft)	massive	40% coarse	1.715	0.353	0.394	4.116	10.45	
3Cg	0.89–0.97	clay	0	massive	prominent redox	1.560	0.411	ND	ND	ND	
4Cg	0.97–1.37	sandy loam	90 (soft)	massive	5% medium prominent redox and concentrations	1.673	0.369	0.060	0.186	3.08	
5Cg	1.37–1.43	clay	0	massive	40% coarse prominent redox	1.597	0.397	ND	ND	ND	
6Cg	1.43–1.47 +		90 (soft)	massive							

[†] Pedality is described using ped grade, ped size, and ped shape: 1, 2, 3 represent weak, moderate, and strong ped grades, respectively; vf, f, m, and c represent very fine, fine, medium, and coarse ped sizes, respectively; gr, abk, sbk represent granular, angular blocky, and subangular blocky ped shapes, respectively. Massive means no soil structure developed.

‡ ND, no data.

skeletal, mixed, active, mesic Lithic Dystrudepts) is shallow, with depth to bedrock generally <0.5 m, while the Berks series (loamy-skeletal, mixed, active, mesic Typic Dystrudepts) generally has a 0.5- to 1-m depth to bedrock. If depth to bedrock is >1 m, landscape position is used to differentiate the Rushtown series (loamy-skeletal over fragmental, mixed, mesic Typic Dystrudepts) located in swale centers from the Ernest series (fine-loamy, mixed, superactive, mesic Aquic Fragiudults) and the Blairton series (fine-loamy, mixed, active, mesic Aquic Hapludults), both located on the valley floor. Presence or absence of a fragipan-like layer (with prismatic structure and prominent redoximorphic features and clay films, but not dense and brittle enough to be called fragipan) and depth to redoximorphic features are then used to further separate the Ernest series (with many redox features and a fragipan-like layer starting at the 0.3–0.5 m depth) from the Blairton series (with few redox features starting at ~1-m depth). For more details on the soil survey and mapping conducted in this catchment, readers are referred to Lin et al. (2006a).

Five representative soil pits, one for each of the five soil series mapped, were excavated and fully described in situ and then sampled for laboratory characterizations. A 0.06-m-high, 0.054-m-diameter brass ring was used to collect intact soil cores (in triplicates) for each soil horizon in both vertical and lateral orientations. A soil pit was excavated to collect these samples. Briefly, the brass rings were inserted laterally at desired soil depths from a freshly dug trench wall to collect horizontally oriented soil cores. The vertically oriented soil samples were collected by digging downward to desired depths and then inserting the rings into the soil vertically. Saturated hydraulic conductivity (K_{sat}) for both lateral and vertical orientations was determined in the laboratory using the constant head method (Klute, 1986). After the K_{sat} measurements, soil bulk density was determined for each soil core, from which total soil porosity was calculated (Table 1).

Soil Moisture and Hydrologic Monitoring and In Situ Observations

Based on the detailed mapping of landforms and soils in the catchment, 77 sites were selected to monitor soil moisture at multiple depths throughout the Shale Hills (Fig. 2). These 77 sites covered all landforms and soil types in the catchment (Table 2), with many of them aligned in transects (Fig. 2). Using the stream and an extended dash line indicated in Fig. 2, these 77 sites were separated into south- and north-facing sites and then further grouped based on the four landform units (hilltop, hillslope, swale, and valley floor) (Table 2).

A V-notch weir equipped with a continuous water level recorder was used to monitor stream discharge at the outlet of the catchment. Stream stage data were collected every 15 min and converted to discharge using a rating curve; data were integrated to daily stream discharge for the monitoring period. Daily precipitation was recorded in a weather station 0.8 km away from the study catchment. In addition, 16 tipping-bucket rain gages with HOBO recorders (Davis Instruments Corp., Hayward, CA) were placed throughout the catchment (Table 2) to monitor rainfall variability from 29 Sept. 2004 until 8 Dec. 2004. These gages record all rainfall at a height of 0.3 m above the ground; each tip represents 0.2 mm of rain.

The soil moisture monitoring was conducted from 24 Mar. 2004 to 19 Mar. 2005 (Fig. 3). In Phase I of this study, 30 sites (Table 2) were initially monitored from 24 Mar. to 15 Aug. 2004. An additional 47 sites (Table 2) were installed in Phase II of this study, and their monitoring began on 10 Sept. 2004. Generally speaking, twice weekly measurements were made from 24 Mar. to 7 June 2004, and daily measurements were

Table 2. Basic characteristics of the 77 monitoring sites at the Shale Hills Catchment.

Landform	Soil series	Phase II site ID	Phase I site ID	Subsoil wetness cluster†	TDR access tube depth	Soil solum thickness	Local slope	DEM-derived slope	Topographic wetness index	Elevation	Rain gage placement
					cm		%	m			
North-facing slope sites											
Hilltop	Weikert	3	MD6	D1	45	33	13.4	12.6	6.07	278.61	Yes
Hilltop	Weikert	14		D1	30	14	9.0	12.3	6.53	283.35	
Hilltop	Weikert	24	MD14	D1	44	30	14.8	24.0	5.50	295.16	
Hilltop	Weikert	30		D1	48	30	7.4	21.5	4.89	297.62	Yes
Hilltop	Weikert	36		D1	58	24	41.0	24.9	5.72	303.27	
Hillslope	Weikert	2	MD5	D2	83	53	31.8	36.1	5.92	270.74	
Hillslope	Weikert	4	MD5A	D1	85	30	29.0	34.7	5.93	266.43	Yes
Hillslope	Weikert	5	MD5B	D2	76	53	29.7	29.7	6.01	274.12	
Hillslope	Weikert	7		D2	68	30	47.5	32.2	6.08	272.05	
Hillslope	Weikert	8		D1	94	36	47.5	32.5	6.14	271.14	Yes
Hillslope	Weikert	9		D1	109	27	41.0	32.1	5.98	273.16	
Hillslope	Weikert	10		D1	71	29	36.9	30.7	5.97	274.11	
Hillslope	Weikert	23	MD13	D1	99	40	38.1	38.6	6.36	280.69	Yes
Hillslope	Weikert	25	MD13A	D1	93	40	31.2	30.8	6.85	274.72	
Hillslope	Weikert	26	MD13B	D1	63	23	36.7	35.7	6.11	286.72	
Hillslope	Weikert	29		D1	56	65	21.3	31.1	7.30	284.66	Yes
Hillslope	Weikert	31		D1	101	23	41.0	31.1	6.15	283.91	
Hillslope	Weikert	33	MD16A	D1	101	38	31.2	20.2	8.75	284.17	
Hillslope	Weikert	34		D2	70	50	36.1	23.6	7.57	287.67	Yes
Hillslope	Weikert	35	MD17	D1	109	66	25.0	32.4	6.23	295.30	
Hillslope	Weikert	A1		D2	108	55	41.0	28.2	6.74	271.71	
Hillslope	Weikert	A2		D2	79	41	47.5	29.1	6.69	272.22	Yes
Hillslope	Weikert	A3		D2	63	35	47.5	30.2	6.59	274.23	
Hillslope	Weikert	A4		D1	73	26	41.0	27.8	6.55	277.19	
Hillslope	Weikert	A5		D1	53	21	24.6	26.6	6.21	283.41	Yes
Hillslope	Weikert	B1		D1	113	50	41.0	28.3	6.70	271.94	
Hillslope	Weikert	B2		D1	63	39	32.0	34.3	6.23	279.57	
Hillslope	Weikert	B3		D1	109	37	32.8	34.2	6.11	283.05	Yes
Hillslope	Weikert	B4		D1	71	36	27.9	33.0	6.05	285.11	
Hillslope	Weikert	B5		D1	40	30	34.4	31.8	6.03	286.44	
Swale	Berks	13		D2	87	87	18.0	23.0	6.66	280.36	Yes
Swale	Berks	28		W1	98	90	23.0	12.9	9.24	272.69	
Swale	Rushtown	12		W2	107	212	27.9	25.9	8.29	269.79	
Valley floor	Ernest	1	MD4	W1	91	152	18.6	4.4	10.06	261.49	Yes
Valley floor	Ernest	6		W1	109	140	9.0	2.9	13.88	262.93	
Valley floor	Ernest	11	MD7	W1	103	152	16.5	7.9	9.90	264.66	
Valley floor	Ernest	15		W1	97	150	6.6	2.9	12.75	268.69	Yes
Valley floor	Ernest	22	MD12	W1	95	152	15.7	6.2	11.31	269.96	
Valley floor	Ernest	27		W1	107	140	12.3	10.4	9.40	271.07	
Valley floor	Rushtown	32	MD16	W1	113	152	6.2	18.1	8.49	281.53	Yes
South-facing slope sites											
Hilltop	Weikert	44	MD2B	D1	95	38	15.3	18.3	6.14	277.58	Yes
Hilltop	Weikert	49	MD1	D1	59	35	16.2	18.4	5.52	287.02	
Hilltop	Weikert	54	MD11	D1	30	30	18.5	10.0	5.76	294.24	
Hilltop	Weikert	67		D1	35	10	4.9	7.7	5.67	296.86	Yes
Hillslope	Weikert	40		D1	43	22	19.7	17.5	6.33	269.55	
Hillslope	Weikert	42	MD2A	D1	49	30	24.3	24.5	5.95	274.26	
Hillslope	Weikert	43	MD2	D1	35	30	24.4	18.3	5.99	276.93	Yes
Hillslope	Weikert	47		D2	71	38	24.6	17.3	6.20	269.25	
Hillslope	Weikert	58	MD9B	D1	20	25	20.5	33.1	5.83	280.32	
Hillslope	Weikert	59		D1	29	20	23.8	29.5	5.60	292.76	Yes
Hillslope	Weikert	60		D1	34	20	29.5	30.9	6.25	276.62	
Swale	Berks	46		D2	66	45	28.7	20.1	7.25	271.31	
Swale	Berks	48		W1	110	145	34.4	29.7	6.32	276.72	Yes
Swale	Berks	53	MD10	W1	108	104	38.4	34.0	6.02	289.79	
Swale	Berks	55		W1	112	110	27.9	23.8	7.47	273.47	
Swale	Berks	56		W1	113	301	32.0	32.3	7.19	278.20	Yes
Swale	Berks	57	MD9A	D2	100	60	30.5	31.6	6.75	278.71	
Swale	Berks	68		D2	65	65	32.8	36.2	5.70	279.94	
Swale	Berks	71		D2	79	79	38.5	27.7	5.88	291.17	Yes
Swale	Rushtown	41		W2	113	300	31.1	24.9	6.62	268.22	
Swale	Rushtown	45		W2	113	167	23.8	20.6	7.46	272.08	
Swale	Rushtown	50	MD8	W2	113	377	15.1	16.6	8.91	271.55	Yes
Swale	Rushtown	51		W2	113	300	13.1	19.2	8.73	272.63	
Swale	Rushtown	52	MD9	W1	111	152	22.7	27.9	7.94	277.45	
Swale	Rushtown	65		W2	113	300	32.8	32.1	6.90	281.01	Yes
Swale	Rushtown	66		W2	107	300	43.4	36.3	5.94	290.21	
Swale	Rushtown	69	MD16B	W2	107	109	31.9	25.9	7.01	282.03	
Swale	Rushtown	70	MD18	W2	112	185	26.4	25.8	5.90	290.03	Yes
Swale	Rushtown	72	MD20	W2	91	193	20.6	26.7	5.53	294.92	
Swale	Rushtown	73	MD19	W2	111	114	30.2	22.3	5.92	292.33	
Valley floor	Berks	38		W1	111	145	19.7	13.5	6.66	262.75	Yes
Valley floor	Blairton	61		W2	110	246	10.7	8.8	8.52	274.19	

Continued next page.

Table 2. Continued.

Landform	Soil series	Phase II site ID	Phase I site ID	Subsoil wetness cluster†	TDR access tube depth	Soil solum thickness	Local slope	DEM-derived slope	Topographic wetness index	Elevation	Rain gage placement
					cm		%			m	
Valley floor	Blairton	62	MD15	W1	110	204	9.8	9.9	8.46	273.33	
Valley floor	Blairton	63		W1	110	156	6.2	7.5	11.18	275.18	
Valley floor	Blairton	64		W1	113	204	11.5	7.6	12.47	274.56	
Valley floor	Ernest	37	MD3	W1	107	140	8.2	8.8	7.18	261.90	
Valley floor	Ernest	39		W1	100	102	31.2	20.4	6.48	264.57	

† Subsoil wetness clusters were based on a combined consideration of soil thickness, topographic wetness index, and local slope. D1, dry; D2, moderately dry; W2, moderately wet; W1, wet.

conducted from 14–30 June 2004 and from 13 July to 15 Aug. 2004. Approximately weekly data were collected from 10 Sept. to 18 Dec. 2004 and from 13 Feb. to 19 Mar. 2005. From mid December 2004 to mid February 2005, no monitoring data were collected because of snow cover and frozen soils.

Surface soil moisture measurements were performed using a Theta probe combined with a HH2 readout (Delta-T Devices, Burwell, UK), which uses dielectric technology to determine volumetric water content. Each monitoring site had a specific area designated to insert the probe. Leaf litter (Oe horizon) was removed before taking measurements and then replaced after five replicate measurements were taken. The average of the five replicates was used in the analysis. The Theta probe was designed to take moisture measurements to a

soil depth of 0.06 m. Multidepth subsurface soil moisture measurements were performed using a TRIME-T3 tube access probe and a mobile moisture meter TRIME-FM3 (IMKO, Ettlingen, Germany). This instrument uses time domain reflectometry (TDR) technology and was designed to take volumetric soil moisture readings while being placed at desired depth inside a 0.051-m-diameter Schedule 40 PVC access tube. The effective signal penetration depth of the probe is about 0.15 m, with the highest sensitivity in the immediate vicinity of the access tube decreasing exponentially with distance (IMKO, 2000). The electric field distribution of the probe is elliptical; thus a higher representation of soil moisture can be achieved by rotating the probe to take multiple measurements and calculating the average. In the Phase I of this study, three sets

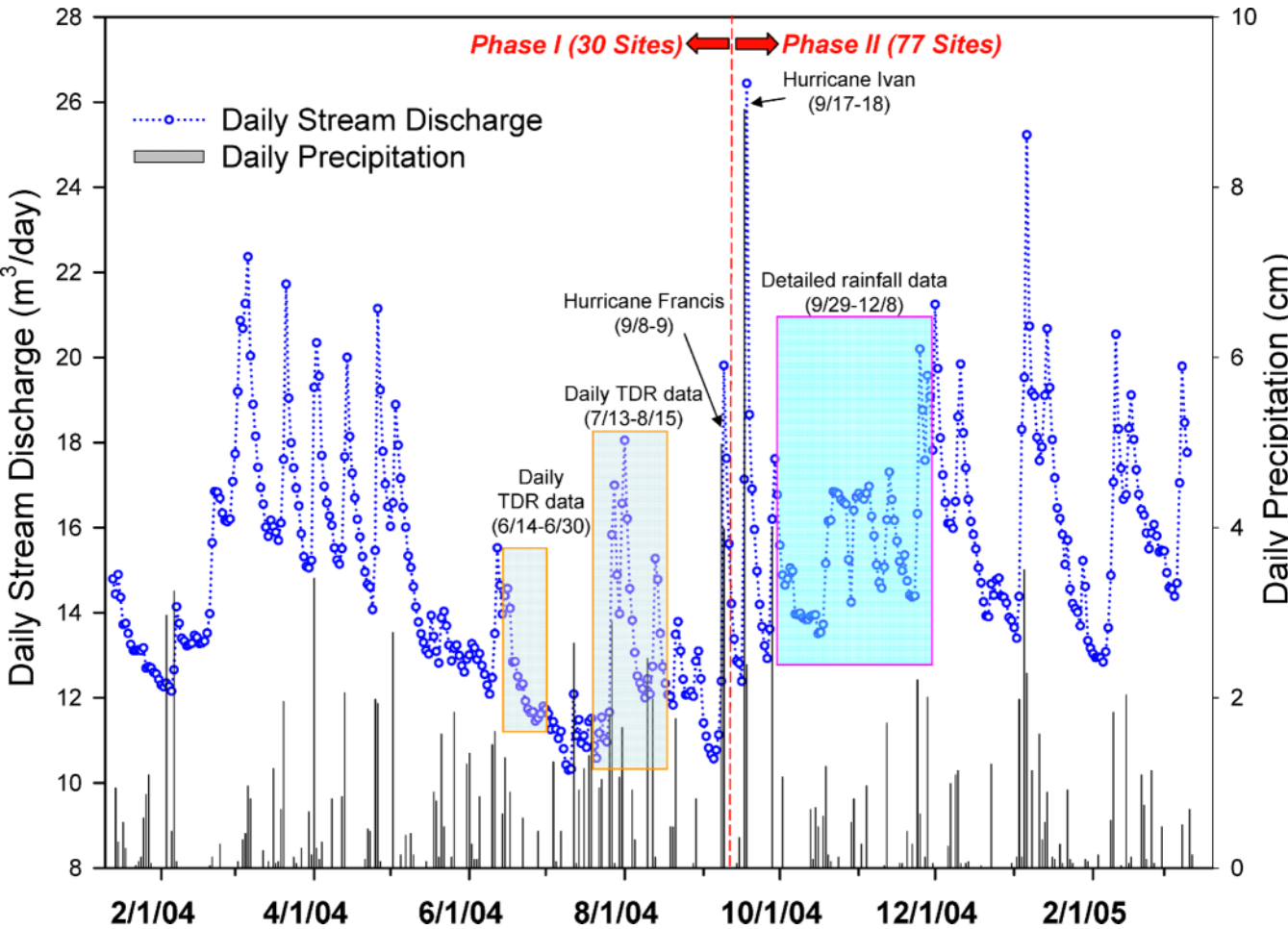


Fig. 3. Daily precipitation and stream discharge at the outlet of the catchment for the period of this study (24 Mar. 2004–19 Mar. 2005). Daily soil moisture data were collected for two periods from 14 to 30 June and from 13 July to 15 Aug. 2004. Detailed precipitation data were collected using 16 rain gauges from 29 Sept. to 8 Dec. 2004.

of measurements were taken at each depth, with the probe rotated one-third of a turn between each set. In Phase II, two sets of measurements were taken with the probe rotated 90° because the replicates have consistently shown fairly similar readings. Besides, the larger number of monitoring sites required a shorter measurement time per site. Nevertheless, if any questionable reading was noticed, a third reading was taken to ensure representation. Measuring accuracy of the TRIME-T3 probe is $\pm 2.0\%$ in the range of 0 to 40% and $\pm 3.0\%$ in the range of 40 to 60% (IMKO, 2000).

The TDR access tubes were installed to a maximum depth of 1.1 m (because of the length limitation of the Giddings Auger Kit used for installation). The bulk of the soil was removed using a tapered bit on a metal pipe that was pounded into the ground with a slide hammer. A second metal tube was then used with an inverse cutting bit that shaped the hole to slightly smaller than 0.051-m diameter to ensure a tight fit of the PVC tube against the soil. The bottom end of the access tube was capped with a PVC cemented test cap. The tube was then placed into the augured hole with a tight fit with the surrounding soil. The top end was capped with a removable PVC end cap. After installation, the access tubes were left undisturbed for several months so settling could occur, resulting in good tube–soil contact. Actual measurements were taken by lowering the TRIME-T3 probe into the access tube with the waveguides fitting tightly against PVC pipe walls. Readings were then taken at three depth intervals of 0.11 to 0.29, 0.51 to 0.69, and 0.91 to 1.09 m (representing the 0.2-, 0.6-, and 1.0-m depths, respectively) during the monitoring period from 24 Mar. to 21 July 2004. Starting 22 July 2004, measurements were collected at every 0.2-m depth interval down to 1.1 m (i.e., the 0.2-, 0.4-, 0.6-, 0.8-, and 1.0-m representative depths, respectively) to calculate soil profile moisture storage. The TRIME-T3 probe is 0.2 m long (with an effective measuring length of 0.18 m), and its midpoint was used to determine the measurement depth from the soil surface (e.g., 0.2 m from the soil surface to the probe midpoint reflects 0.11–0.29 m soil depth being measured). In addition, we also took the reading for the 0.01- to 0.19-m soil depth (i.e., the 0.1-m depth) so the soil depth of 0 to 0.10 m (i.e., no overlap with the next reading of 0.11–0.29 m) could be included in the solum water storage calculation using the same measurement methodology. If the access tube at a site did not allow for measurements down to 1.1 m because of shallower depth to bedrock, the last measurement was taken at the bottom of the access tube, which is labeled as the “deepest measurement depth” in this paper. Table 2 provides the specific depth for all 77 access tubes installed (i.e., the deepest measurement depth).

The measured volumetric soil moisture contents at 0.2-m depth intervals were then used to calculate soil profile moisture storage within the top 1.1-m solum (i.e., A, E, and B horizons), S (water depth per unit area):

$$S = \sum_{z=0}^b \theta_z d_z \quad [1]$$

where θ_z is volumetric soil moisture content measured by the TRIME-T3 probe at the 0.2-m depth interval, d_z is soil depth interval (0.2 m in this study) down to a depth of b , which is 1.1 m (maximum monitoring depth) or the bottom of the solum if it is shallower than 1.1 m. Soil solum thickness was determined for each site during soil mapping and instrument installation, where a soil core for each site was collected and examined (Table 2).

For each monitoring site, we also measured local slope and calculated terrain attributes. A meter stick was placed on the land surface and oriented in the steepest downslope direction.

The downhill end of the stick was raised until level (using a leveling tool), and the distance from the downhill end of the stick to the ground was recorded. The local slope is the ratio of rise over run. The local slopes measured in situ and the slope values derived from the refined 3-m DEM show a nearly linear relationship (local slope = $1.087 \times$ DEM-derived slope), with a $R^2 = 0.5659$ (Table 2). The refined DEM was then used to calculate terrain attributes for each monitoring site using the ArcGIS software (ESRI, Redlands, CA). Particularly relevant terrain attributes used in this paper included slope and topographic wetness index (TWI). The TWI is defined as:

$$TWI = \ln(a/\tan\beta) \quad [2]$$

where a is the upslope contributing area calculated with the D-inf algorithm described in Tarboton (1997) and $\tan\beta$ is the slope calculated as the steepest outwards slope on one of eight triangular facets centered at each grid cell, measured as drop/distance (i.e., tan of the slope angle) (Tarboton, 1997).

Furthermore, numerous in situ observations were conducted throughout this study, with a focus on identifying flow pathways in the catchment and their relations to soil characteristics and landscape features. Real-time observations and special data (e.g., video clips, photographs, field notes, and supplemental soil moisture data) were obtained, particularly during periods of snow melt and large storm events that included both Hurricane Francis and Ivan occurred in September 2004. These real-time *direct* observations, while for the most part qualitative in nature, provided valuable evidence of landscape water flow pathways in this catchment.

Data Analysis

To quantify the temporal stability of soil moisture spatial pattern across the study catchment, and to relate the observed temporal stability to soil moisture and hydrologic dynamics, the following two analyses were performed. All statistical analyses in this study were performed using the SAS (SAS Institute Inc., Cary, NC).

Ranking Stability

An approach originally proposed by Vachaud et al. (1985) and later applied by others (e.g., Mohanty and Skaggs, 2001; Martinez-Fernandez and Ceballos, 2003; Grant et al., 2004) was used to compare different monitoring sites relevant to each other and to the whole catchment's mean. The time average ($\bar{\delta}_i$) of relative difference in soil moisture (δ_{ij}) for each monitoring site i at the same measurement depth was calculated as

$$\bar{\delta}_i = \frac{1}{m} \sum_{j=1}^m \delta_{ij} \quad [3]$$

$$\delta_{ij} = \frac{\theta_{ij} - \bar{\theta}_j}{\bar{\theta}_j} \quad [4]$$

where m is the number of monitoring days, θ_{ij} is the soil moisture content of a given depth at site i on measurement day j , and $\bar{\theta}_j$ is the average moisture content at a given depth on measurement day j for all monitoring sites across the catchment:

$$\bar{\theta}_j = \frac{1}{N} \sum_{i=1}^N \theta_{ij} \quad [5]$$

where N is the total number of monitoring sites ($N = 30$ in Phase I and $N = 77$ in Phase II of this study). Positive or negative values of $\bar{\delta}_i$ suggest overestimates or underestimates of the catchment average moisture content regardless of

measurement time. Vachaud et al. (1985) suggested that the variation associated with δ_{ij} , called temporal standard deviation $\sigma(\delta_{ij})$, is an indicator of temporal stability. Low value of $\sigma(\delta_{ij})$ has been used to define time stable locations (e.g., Vachaud et al., 1985; Martinez-Fernandez and Ceballos, 2003).

Pachepsky et al. (2005) used an essentially same concept to test temporal stability. They calculated relative moisture content β_{ij} at site i at measurement time j instead of relative difference δ_{ij} :

$$\beta_{ij} = \frac{\theta_{ij}}{\theta_j} \quad [6]$$

The value of β_{ij} equals to $\delta_{ij} + 1$. Values of β_{ij} greater than or less than 1 indicate wetter or drier than overall average moisture content in the entire catchment. Pachepsky et al. (2005) stated that the larger the standard deviation of the relative moisture content $\sigma(\beta_{ij})$, the less temporal stability.

The δ_{ij} with $\sigma(\delta_{ij})$ or β_{ij} with $\sigma(\beta_{ij})$ evaluate the ranking stability among different monitoring sites within a study area. By plotting these values according to relative ranking, this analysis determines whether certain sites are consistently indicative of catchment characteristics (such as catchment average moisture content, or the driest or wettest conditions) regardless of the observation time.

Another test commonly used to determine relative ranking stability among different sites is the nonparametric Spearman rank correlation coefficient (r_s), calculated as (Vachaud et al., 1985)

$$r_s = 1 - \frac{6 \sum_{i=1}^n [R_2(i) - R_1(i)]^2}{n(n^2 - 1)} \quad [7]$$

where n is the number of spatial data points, $R_1(i)$ is the rank of soil moisture at site i among all the monitoring sites measured at two successive times (t_1 and t_2), and r_s gauges the ranking observed at t_2 described by t_1 . The closer r_s is to 1, the more stable the process involved.

Slope-Intercept Analysis of Linear Regression

Kachanoski and de Jong (1988) suggested that linear regression and simple correlation between successive time intervals would be suffice as indicators of temporal stability across all spatial locations. Soil moisture content θ at site i at two times (t_1 and t_2) can be regressed as:

$$\theta_{t_2}(i) = a_{t_2-t_1} \theta_{t_1}(i) + b_{t_2-t_1} + \epsilon_{t_2-t_1} \quad [8]$$

where $a_{t_2-t_1}$, $b_{t_2-t_1}$, and $\epsilon_{t_2-t_1}$ are the regression slope, intercept, and error, respectively. This linear regression analysis can

avoid the possible contradictory results in the two time stability tests suggested by Vachaud et al. (1985), where addition of a constant amount of water to each spatial location could result in the relative difference δ_{ij} not remaining the same (Kachanoski and de Jong, 1988). The linear regression parameters of a (slope) and b (intercept) in Eq. [8] were used in this study to further examine temporal stability in relation to soil moisture and hydrologic dynamics using the approach suggested by Grant et al. (2004).

Pearson correlation coefficient (r_p) was used to evaluate the degree of linear correlation, which is given by (Kachanoski and de Jong, 1988)

$$r_p = \frac{\text{Cov}[\theta_{t_2}(i), \theta_{t_1}(i)]}{(\text{Var}[\theta_{t_2}(i)] \text{Var}[\theta_{t_1}(i)])^{1/2}} \quad [9]$$

where $\text{Cov}[]$ and $\text{Var}[]$ are the covariance and variance operators, respectively. The value of r_p indicates how much of the spatial variance observed at t_2 can be explained by the variation represented at t_1 , that is, the persistence of the spatial pattern across all locations (space average). Values of r_s (rank) and r_p (absolute value) are not necessarily correlated (Grant et al., 2004).

Given that both r_s and r_p are statistically significant (which was the case for nearly 100% in this study), four temporal stability conditions can be described using various combinations of a and b (Grant et al., 2004). A two-tailed t test, with $\alpha = 0.05$, was used to test whether b is significantly different from zero, and an F test was used to test whether a is significantly different from one (SAS, 1999). On the basis of this, four scenarios of soil moisture and hydrologic dynamics leading to temporal stability could be interpreted, as summarized in Table 3. Condition 1 (no change) occurs if a and b are not significantly different from one and zero, respectively. This condition implies that there is effectively no net change in soil moisture between the two dates compared. Condition 2 (uniform change) is characterized by b significantly different from zero while a is not significantly different from one, suggesting soil moisture change is approximately equal across all spatial locations. Condition 3 (nonuniform change) applies when b is zero but a is significantly different from one. In this case, change in soil moisture is not evenly distributed between measurement dates, but rather, some sites change more than others. The fourth condition (complex change) refers to the scenario where a is significantly different from one and b is significantly different from zero, indicating nonuniform change caused by variable inputs–outputs across spatial locations or the occurrence of complex water redistribution. Grant et al. (2004) noted that, for each of these four conditions, the pattern of ranks and values are temporally persistent, but the mean and/or variation among values may change.

Table 3. Four temporal stability conditions based on the slope-intercept analysis of linear regression (modified from Grant et al., 2004).

Temporal stability condition	Slope (a) significantly different from 1	Intercept (b) significantly different from 0	Interpretations
Condition 1 (No change)	No	No	Water contents at different times are nearly identical; may be due to relatively little input and/or output, or output at each site equals input, or to an extended period that buffers catchment response.
Condition 2 (Uniform change)	No	Yes	Nearly uniform change in soil water pattern over catchment; may be caused by uniform input (rain) and/or output (plant water uptake, shallow evaporation and/or drainage).
Condition 3 (Nonuniform change)	Yes	No	Nonuniform change with time; may be due to a lack of uniform input/output while some locations are drying or wetting more than others or redistribution occurs.
Condition 4 (Complex change)	Yes	Yes	Complex nonuniform change; caused by nonuniform input (rain-on-snow or snowmelt) or output (transpiration) or a combination of uniform input/output and redistribution.

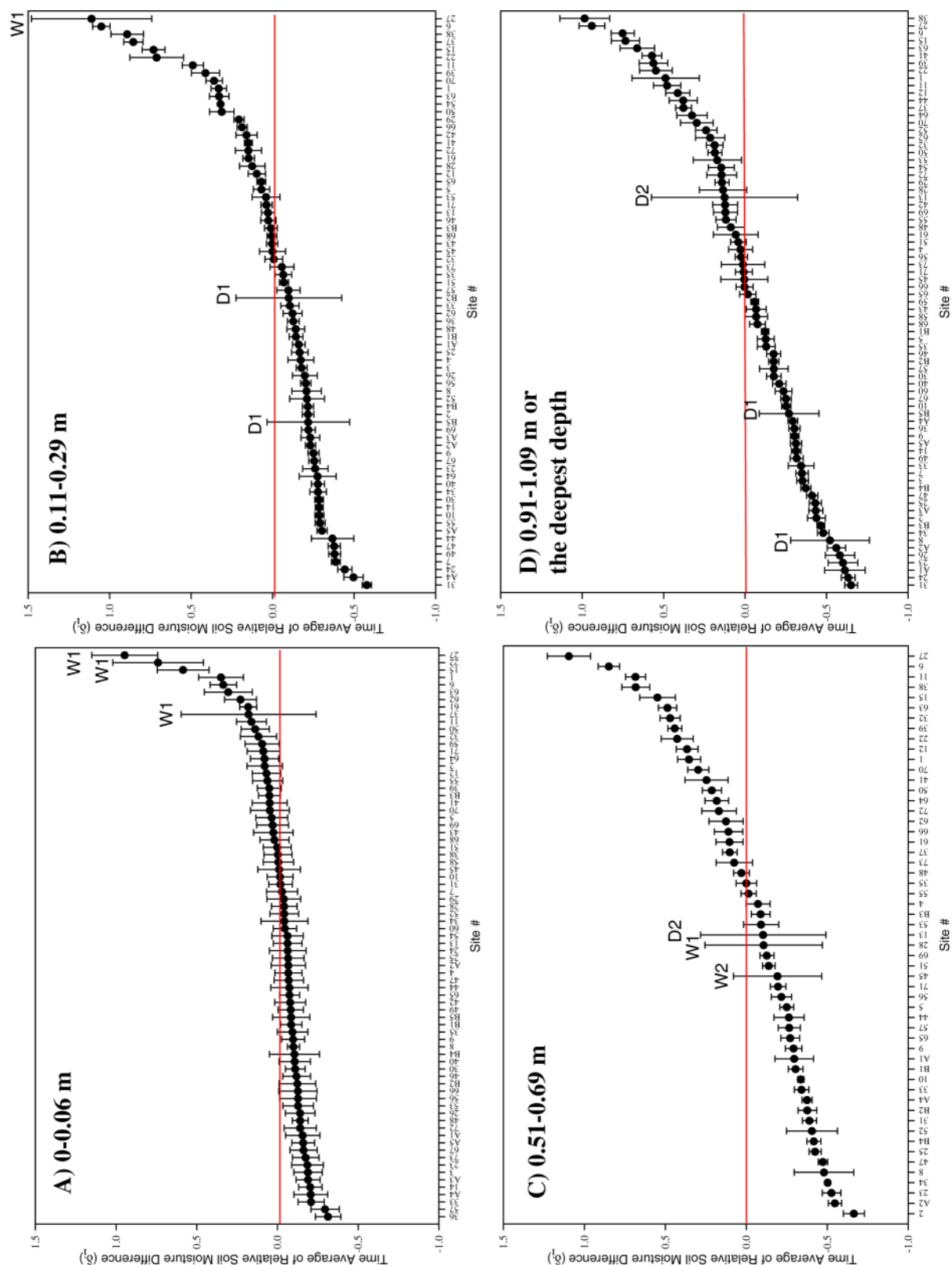


Fig. 4. Time average plots of ranked relative soil moisture content deviated from the catchment-wide mean value at four monitoring depths. Vertical bars correspond to the associated temporal standard deviation for the 1-yr monitoring period. Site numbers refer to the monitoring locations indicated in Fig. 2 and Table 2. A few sites with relatively large temporal standard deviation are identified using soil moisture clusters of dry (D1), moderately dry (D2), moderately wet (W2), or wet (W1) groups.

In addition to the above two analyses, a cluster analysis suggested by Lin et al. (2006a) was used to group the 77 monitoring sites into four clusters (wet, moderately wet, moderately dry, and dry) (Table 2). This cluster analysis was based on a combined consideration of depth to bedrock (i.e., soil depth, a key feature used to differentiate the five soil series in the catchment), topographic wetness index (Eq. [2]), and local slope. Lin et al. (2006a) found that these three attributes provided a good interpretation of observed subsoil moisture patterns in the Shale Hills Catchment. Sites in a given cluster are believed to be similar to each other in some sense (as suggested by the data, not defined a priori), and sites in different clusters tend to be dissimilar.

RESULTS AND DISCUSSION

Temporal Stability of Soil Moisture Spatial Pattern as Governed by Soil Types and Landforms

The time average plots of relative soil moisture difference (δ_i) across the catchment at multiple depths showed a strong temporal stability in the ranks of the monitoring sites (Fig. 4). The temporal standard deviation $\sigma(\delta_i)$ for the majority of the 77 sites was 9.5% ($\pm 2.3\%$), 5.4% ($\pm 2.7\%$), 7.0% ($\pm 3.5\%$), and 6.7% ($\pm 3.7\%$) for the 0- to 0.06-, 0.11- to 0.29-, 0.51- to 0.69-m, and the deepest (≤ 0.91 –1.09 m) monitoring depths, respectively, excluding a few sites with relatively large $\sigma(\delta_i)$ as identified in Fig. 4. These sites with relatively large $\sigma(\delta_i)$ were associated with wet locations at the surface and mostly dry sites at the subsurface. Monitoring sites along the valley floor (e.g., Sites 6, 27, and 38) were consistently wetter than the catchment average throughout the soil profiles (by as much as $\approx 100\%$ greater in volumetric moisture content), while the sites at the convex hillslope (e.g., Sites 24, 31, and 36)

were much drier than the catchment average in much of the soil profiles regardless of the measurement time (>30 and 50% lower in soil volumetric moisture content for the surface and subsurface, respectively) (Fig. 4). The sites representing the catchment average soil moisture content within $<1\%$ of the overall time average $\bar{s}_i$ at any time during the monitoring year included Sites 38, 45, 51, and 58 for the 0- to 0.06-m depth (temporal standard deviation being ± 8.7 – 13.2%); Sites 32, 43, 45, 68, and B1 for the 0.11- to 0.29-m depth (deviating by ± 3.0 – 8.0%); Sites 35 for the 0.51- to 0.69-m depth (deviating by $\pm 6.3\%$); and Sites 45, 66, and 71 for the 0.91- to 1.09-m (or the deepest) monitoring depth (deviating by ± 5.2 – 14.4%). The only common site among these locations representing catchment average for different depths was Site 45, with the exception of the 0.51- to 0.69-m depth (where Site 45 was $19.3 \pm 27.3\%$ lower in volumetric moisture content than the catchment average of that depth). Site 45 is located at the middle of a swale that contains a Rushtown soil series, has a subsoil moisture cluster of moderately wet (W2), a topographic wetness index of 7.46, and a local slope of 23.8%.

The relative soil moisture content β_{ij} was greater or less than 1 in the vast majority of the cases in this study (Table 4), again demonstrating considerable temporal stability of soil moisture spatial pattern in this catchment. The Ernest series maintained a mean β_{ij} value of 1.31 to 1.72 at various depths throughout the monitoring year, while the Blairton, Rushtown, Berks, and Weikert series had mean β_{ij} values of 0.99 to 1.70, 0.96 to 1.21, 0.76 to 1.16, and 0.65 to 0.93, respectively, which varied spatially, vertically, and seasonally (Table 4).

The ranking stability among the 77 sites was slightly stronger in the subsurface (5.4–7.0% temporal standard

Table 4. Statistics of relative difference in soil moisture (β_{ij}) in each of the five soil series at the Shale Hills Catchment. Values of β_{ij} greater or less than 1 indicate wetter or drier than the overall average of the entire catchment.

Soil Series	0–0.06 m depth				0.11–0.29 m depth				0.51–0.69 m depth				0.91–1.09 m OR deepest measurement depth			
	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.	Mean	SD	Min.	Max.
Spring																
Ernest	1.38	0.41	0.99	2.37	1.51	0.17	1.25	1.82	1.46	0.16	1.21	1.75	1.45	0.07	1.26	1.62
Blairton	1.40	-	1.31	1.58	1.31	-	1.24	1.37	1.45	-	1.40	1.49	1.70	-	1.56	1.82
Rushtown	0.96	0.11	0.71	1.29	1.01	0.26	0.64	1.43	1.10	0.32	0.16	1.50	1.18	0.08	1.02	1.36
Berks	0.78	0.06	0.64	1.02	1.02	0.09	0.88	1.18	0.91	0.14	0.72	1.07	1.09	0.28	0.80	1.34
Weikert	0.92	0.14	0.44	1.28	0.82	0.18	0.41	1.28	-	-	-	-	0.76	0.30	0.28	1.37
Summer																
Ernest	1.31	0.29	0.85	2.21	1.49	0.22	1.20	2.02	1.49	0.14	1.25	1.79	1.53	0.07	1.33	1.75
Blairton	1.30	-	1.08	1.81	1.34	-	1.22	1.53	1.49	-	1.43	1.59	1.62	-	1.45	1.77
Rushtown	1.01	0.16	0.63	1.45	1.04	0.25	0.62	1.54	1.08	0.30	0.16	1.58	1.14	0.13	0.69	1.48
Berks	0.78	0.16	0.57	1.13	0.93	0.11	0.78	1.19	0.76	0.10	0.63	1.01	0.88	0.26	0.61	1.28
Weikert	0.93	0.12	0.58	1.42	0.82	0.18	0.40	1.24	-	-	-	-	0.78	0.31	0.28	1.58
Fall																
Ernest	1.42	0.32	0.99	2.24	1.68	0.31	0.99	2.57	1.55	0.32	1.04	2.20	1.63	0.21	1.30	2.62
Blairton	1.16	0.11	0.97	1.75	1.01	0.27	0.53	1.37	1.24	0.20	1.03	1.69	1.34	0.32	0.94	1.89
Rushtown	0.98	0.12	0.66	1.41	1.07	0.16	0.65	1.41	1.08	0.27	0.05	1.68	1.21	0.19	0.64	1.63
Berks	0.93	0.13	0.59	1.27	1.04	0.30	0.65	2.00	0.97	0.32	0.23	1.84	1.15	0.39	0.74	3.97
Weikert	0.92	0.11	0.50	1.35	0.81	0.20	0.19	1.31	-	-	-	-	0.74	0.25	0.02	1.51
Winter																
Ernest	1.64	0.33	1.18	2.67	1.72	0.32	1.32	2.36	1.53	0.33	1.05	2.10	1.65	0.18	1.33	2.01
Blairton	1.16	0.12	0.97	1.30	0.99	0.26	0.69	1.33	1.17	0.18	1.01	1.48	1.34	0.31	0.87	1.85
Rushtown	1.09	0.18	0.78	1.49	1.09	0.17	0.77	1.45	1.04	0.23	0.66	1.46	1.21	0.19	0.93	1.64
Berks	1.06	0.25	0.76	1.74	1.07	0.31	0.70	1.97	1.00	0.29	0.61	1.70	1.16	0.33	0.48	2.05
Weikert	0.81	0.17	0.35	1.21	0.78	0.17	0.37	1.18	-	-	-	-	0.72	0.24	0.28	1.41

deviation) compared with the surface (9.5% temporal standard deviation). However, the difference in ranking stability among different subsurface depths was not obvious when all monitoring sites are considered together (Fig. 4). By separating them into the five soil series identified in the catchment, a varying temporal stability as a function of depth is noticeable (Table 4). The Ernest series had decreasing standard deviation of relative soil moisture content $\sigma(\beta_{ij})$ with depth throughout the monitoring year, with a higher magnitude of such decreasing trend in the spring and summer. The Weikert series exhibited an opposite trend—increasing $\sigma(\beta_{ij})$ with depth. The Blairton, Berks, and Rushtown series had the highest $\sigma(\beta_{ij})$ at the middle or bottom of the soil profiles. Such varying depth function of temporal persistence may suggest that the most active soil water change occurred at different depths in different soil series in the catchment.

Temporal stability of subsurface soil moisture ranking within a soil profile is also noticeable when absolute soil moisture content is plotted with time (Fig. 5). Except the surface soil moisture that fluctuated most significantly, the ranking of subsoil moisture content as a function of

depth remained largely unchanged within each of the five soil series (Fig. 5). In general, the 0.11- to 0.29-m depth had the lowest moisture content, which gradually increased to the highest at or near the bottom of the soil profile. The Ernest series was an exception to this trend because its wettest layer was at the middle of the soil profile that corresponded to the soil horizon above a clay layer (Fig. 5A, Table 1). The Weikert series also did not display this trend because its shallow depth to bedrock was generally not much deeper than the 0.11- to 0.29-m depth (Fig. 5E).

Regardless of the measurement time during the 1-yr monitoring period, three sites (Sites 13, 29, and 71) were representative of the catchment averaged soil water storage within the 1.1-m solum (Fig. 6). Site 71 is 3.3% ($\pm 2.2\%$) smaller than the catchment average, Site 29 is 6.3% ($\pm 1.7\%$) smaller, whereas Site 13 is 5.8% ($\pm 18.3\%$) greater. Interestingly, these three sites are at the boarder between the wet and dry sites according to the cluster analysis using a combined consideration of depth to bedrock, topographic wetness index (TWI), and local slope. All the sites with soil moisture storage smaller than the catchment average were in dry (D1) or

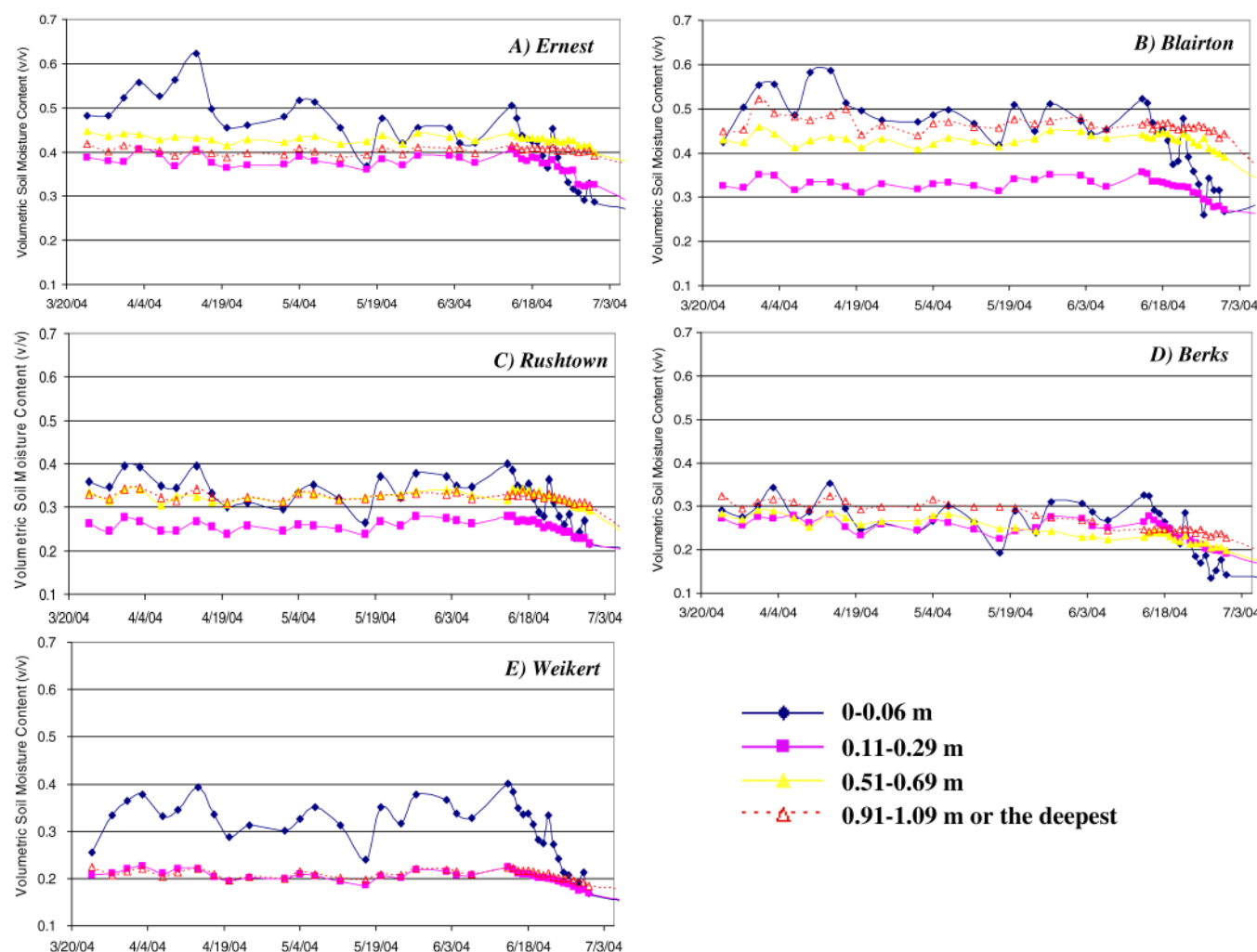


Fig. 5. Temporal change of mean soil volumetric moisture content as a function of depth in each of the five soil series. Note that the lines connecting the measurement points are intended for visualization, which do not represent actual soil moisture change between the measurement dates.

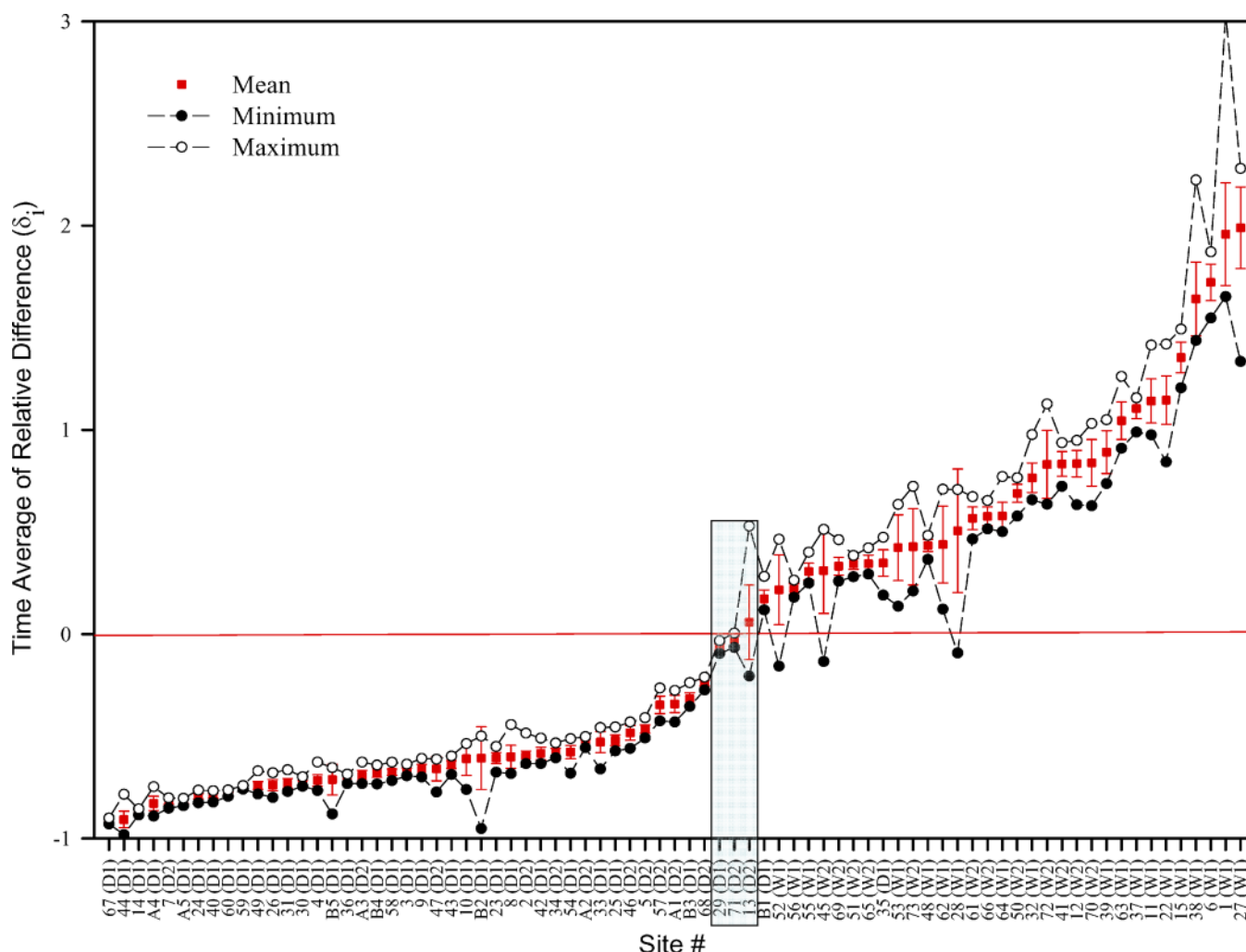


Fig. 6. Time average plot of ranked relative soil moisture storage deviated from the catchment-wide mean. Vertical bars correspond to the associated temporal standard deviation over the 1-yr monitoring period. Solid dots and open circles are minimum and maximum values, respectively. Site numbers refer to the monitoring locations indicated in Fig. 2 and Table 2. Soil moisture clusters of dry (D1), moderately dry (D2), moderately wet (W2), and wet (W1) groups are indicated next to each site number in the parentheses.

moderately dry (D2) clusters, and the sites with moisture storage greater than the catchment average were in wet (W1) or moderately wet (W2) clusters except sites B1 and 35 (Fig. 6). Note that the TWI alone did not provide sufficient differentiation between relatively wet and dry sites in this catchment (Fig. 2 and Table 2). Subsurface features (such as depth to bedrock) must be considered in this regard. The three sites representing the catchment average storage were all near the boundary of a swale, with two mapped as the Berks series and the third one as the Weikert series (Fig. 2). This third site (Site 29) may need to be remapped as a Berks soil since its solum thickness exceeded 0.5 m, the upper limit of the Weikert soil (Table 2). These three sites had solum thicknesses ranging from 0.65 to 0.87 m, local slopes of 23.0 to 31.1%, and TWI values of 5.88 to 7.30 (Table 2). In comparison, Sites 14, 44, and 67 had around 90% smaller soil moisture storage throughout the monitoring year, representing the three driest sites in the catchment (Fig. 6). These three sites were all at the convex hilltop, belonged to the Weikert soil series, and had shallow solum thicknesses (0.10–0.38 m),

relatively small local slopes (4.9–15.3%), and lower TWI values (5.67–6.53) (Fig. 2, Table 2). The wettest three sites in the catchment were Sites 1, 6, and 27, all having nearly 200% greater soil moisture storage within 1.1-m solum than the catchment average (Fig. 6). These three sites were all on the valley floor, belonged to the Ernest soil series, and had >1.4 m solum, 9.0 to 18.6% local slopes, and higher TWI values of 9.40 to 13.88 (Fig. 2, Table 2).

On average, the five soil series identified in the catchment had the following general decreasing trend of moisture storage within the 1.1-m solum throughout the monitoring year: Ernest > Blairton ≥ Rushtown ≥ Berks > Weikert (Fig. 7A). While variation existed depending on landscape location and season, the Ernest soil was generally the wettest in the whole catchment, while the Weikert series was the driest, and the other three soil series fell in between. In this V-shaped catchment, the temporal stability of soil moisture spatial pattern was also governed by landforms, with the following general trend of decreasing moisture storage within 1.1-m solum in both north- and south-facing

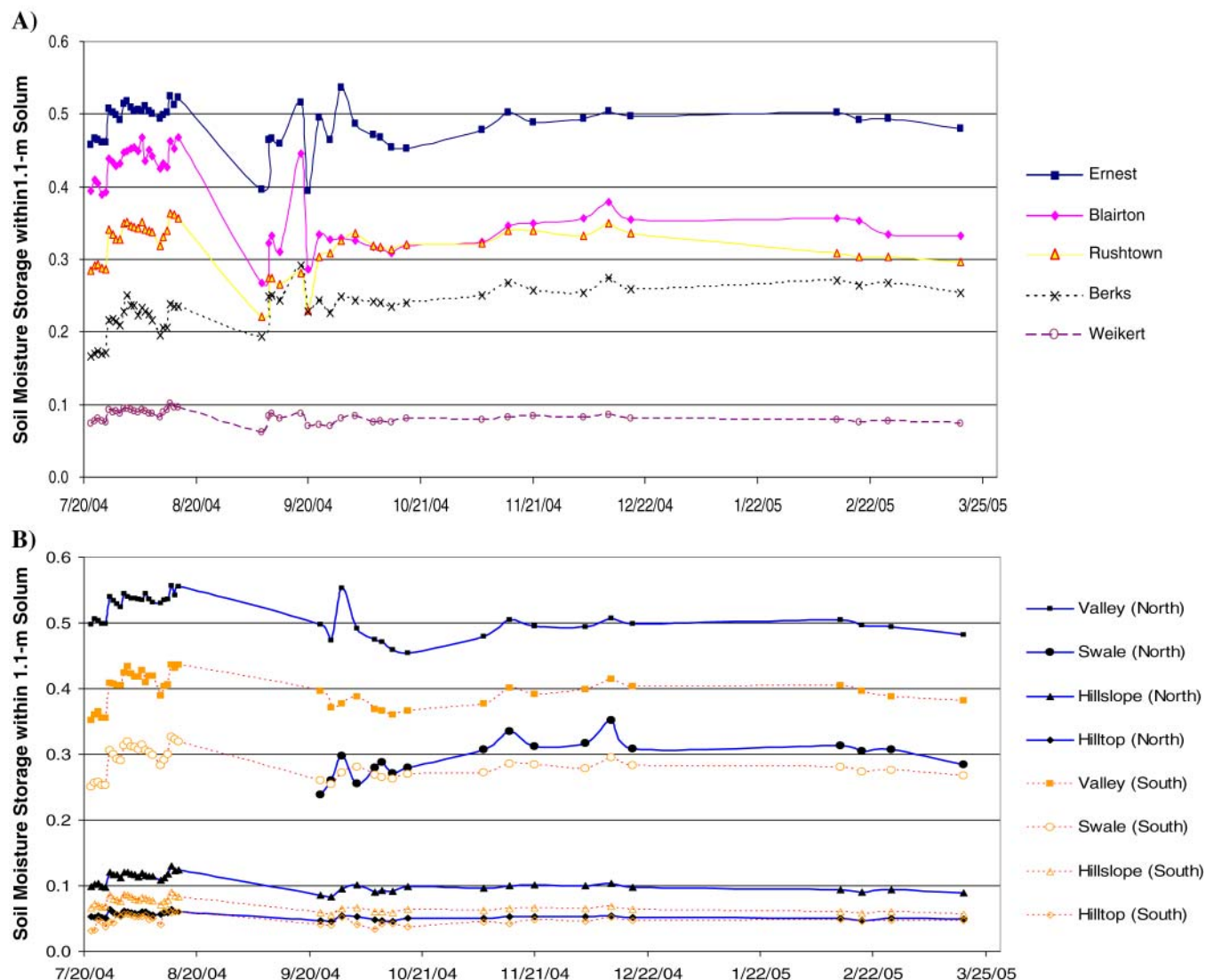


Fig. 7. Averaged soil moisture storage within ≤ 1.1 -m solum from 22 July 2004 to 19 Mar. 2005: (A) grouped by the five soil series and (B) comparison of landform units between the north- and south-facing slopes. Note that the lines connecting the measurement points are intended for visualization, which do not represent actual soil moisture change in between the measurement dates.

slopes: Valley > Swale > Hillslope > Hilltop (Fig. 7B). The north-facing slope stored consistently higher soil moisture within the 1.1-m solum than the south-facing slope in the corresponding landscape positions of valley floor, swale, and hillslope, but the hilltop in both sides of the slopes was similar (Fig. 7B).

Four Temporal Stability Conditions and Soil Moisture Dynamics

Both r_s and r_p were statistically significant for nearly 100% in this study (e.g., Table 5), indicating a high degree of temporal stability of soil moisture spatial pattern in the study catchment. Quantifiable changes of these measures were analyzed using the slope and intercept of Eq. [8] to reveal further information concerning the dynamics of soil moisture and catchment hydrology during the monitoring year. These changes were categorized into four temporal stability conditions

as described in Table 3. The matrices in Table 5 were coded to compare dates and describe the status of spatial soil moisture change with time, while rank remained stable. Such slope-intercept analysis and the related temporal stability conditions can be linked to the soil moisture and hydrologic dynamics in the catchment.

Two time periods shown in Table 5 are representative scenarios—one is from 14 to 30 June and typifies a dry-down period, and the other is from 10 September to 18 December and represents a complex dry-down and wet-up cycle (Table 5, Fig. 3). The first period had daily measurements of soil moisture, and the second had detailed rainfall measurements, both allowing a closer examination of the linkage to soil moisture and hydrologic dynamics. Overall, the June dry-down period had far fewer occurrences of Condition 4 (complex change) compared with the September to December period at various soil depths (Table 5). In the June dry-down period, the surface soil (0–0.06 m) had a good mix of the

Table 5. Matrices of Pearson's correlation coefficient and four conditions of temporal stability of volumetric soil moisture content at multiple depths during two representative monitoring periods.

0–0.06 m																	
	6/14	6/15	6/16	6/17	6/18	6/19	6/20	6/21	6/22	6/23	6/24	6/25	6/26	6/27	6/28	6/29	6/30
6/14	1	§	§	§	§	†	§	†	†	†	†	†	§	‡	§	§	§
6/15	0.82	1	†	†	§	†	§	‡	†	‡	†	†	†	§	§	†	‡
6/16	0.86	0.92	1	†	¶	†	†	†	†	†	†	†	†	†	†	†	‡
6/17	0.85	0.93	0.92	1	†	†	§	†	†	†	†	†	†	†	†	†	‡
6/18	0.88	0.91	0.93	0.91	1	†	†	‡	†	‡	†	†	†	†	‡	‡	†
6/19	0.86	0.94	0.93	0.93	0.94	1	†	†	†	†	†	†	†	†	§	†	†
6/20	0.86	0.85	0.87	0.85	0.90	0.93	1	†	†	†	†	†	†	†	†	†	†
6/21	0.83	0.93	0.91	0.93	0.93	0.92	0.90	1	†	†	†	†	†	†	†	†	†
6/22	0.89	0.86	0.90	0.87	0.91	0.89	0.90	0.89	1	†	†	‡	†	‡	¶	†	¶
6/23	0.82	0.89	0.87	0.89	0.91	0.89	0.90	0.93	0.93	1	†	‡	†	†	§	†	¶
6/24	0.84	0.86	0.87	0.86	0.91	0.89	0.91	0.90	0.91	0.95	1	‡	†	†	§	†	†
6/25	0.81	0.80	0.83	0.82	0.88	0.85	0.89	0.88	0.89	0.95	0.96	1	†	†	§	†	§
6/26	0.63	0.63	0.60	0.65	0.67	0.65	0.62	0.68	0.74	0.77	0.76	0.81	1	†	¶	¶	§
6/27	0.82	0.74	0.75	0.70	0.83	0.78	0.78	0.81	0.85	0.83	0.83	0.85	0.74	1	¶	¶	¶
6/28	0.79	0.80	0.78	0.80	0.83	0.76	0.78	0.84	0.89	0.92	0.89	0.88	0.79	0.85	1	†	†
6/29	0.72	0.79	0.82	0.76	0.86	0.79	0.79	0.83	0.88	0.91	0.89	0.89	0.79	0.83	0.91	1	§
6/30	0.76	0.83	0.83	0.80	0.86	0.81	0.82	0.86	0.90	0.93	0.92	0.91	0.82	0.84	0.94	0.96	1
	9/10	9/12	9/19	9/26	10/3	10/10	10/17	11/17	11/14	11/21	12/5	12/12	12/18				
9/10	1	§	¶	§	¶	¶	¶	¶	¶	¶	†	†	†				
9/12	0.89	1	¶	†	¶	¶	¶	‡	‡	¶	†	§	†				
9/19	0.88	0.90	1	¶	†	†	†	†	‡	‡	‡	¶	¶	†			
9/26	0.86	0.96	0.91	1	¶	¶	¶	†	‡	‡	‡	¶	¶	†			
10/3	0.78	0.93	0.90	0.86	1	†	¶	¶	¶	¶	†	¶	‡				
10/10	0.66	0.88	0.83	0.88	0.87	1	¶	†	†	¶	†	†	†				
10/17	0.68	0.87	0.83	0.80	0.85	0.90	1	¶	¶	¶	†	¶	†				
11/7	0.73	0.90	0.81	0.85	0.89	0.89	0.87	†	¶	†	†	†	†				
11/14	0.77	0.92	0.84	0.86	0.86	0.88	0.87	0.93	1	¶	†	§	†				
11/21	0.69	0.86	0.81	0.81	0.86	0.85	0.84	0.88	0.87	1	†	¶	¶				
12/5	0.74	0.83	0.80	0.82	0.80	0.79	0.75	0.83	0.83	0.83	1	†	†				
12/12	0.89	0.92	0.87	0.85	0.86	0.83	0.84	0.88	0.88	0.82	0.79	1	†				
12/18	0.71	0.85	0.87	0.77	0.78	0.75	0.74	0.80	0.79	0.76	0.76	0.80	1				
0.11–0.29 m																	
	6/14	6/15	6/16	6/17	6/18	6/19	6/20	6/21	6/22	6/23	6/24	6/25	6/26	6/27	6/28	6/29	6/30
6/14	1	†	†	†	†	†	†	§	†	§	§	§	§	§	§	§	§
6/15	0.97	1	§	§	†	†	†	§	†	§	§	§	§	§	§	§	§
6/16	0.97	1.00	1	†	†	†	†	†	†	†	§	§	†	†	§	§	§
6/17	0.96	0.99	0.99	1	†	†	†	†	†	†	§	§	†	†	§	§	§
6/18	0.96	0.99	1.00	0.99	1	†	†	§	†	§	§	§	§	§	§	§	§
6/19	0.96	0.99	0.99	0.99	0.99	1	†	§	†	§	§	§	§	§	§	§	§
6/20	0.94	0.98	0.98	0.98	0.99	0.99	1	†	†	†	§	§	†	†	§	§	§
6/21	0.98	0.98	0.98	0.98	0.99	0.98	0.97	1	†	†	†	†	†	†	§	§	§
6/22	0.96	0.98	0.98	0.98	0.99	0.99	0.99	0.99	1	§	§	§	†	†	§	§	§
6/23	0.96	0.98	0.98	0.98	0.99	0.99	0.99	0.99	1.00	1	§	†	†	†	§	§	§
6/24	0.96	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.99	1.00	1	†	†	†	§	§	§
6/25	0.95	0.97	0.98	0.98	0.99	0.99	0.99	0.98	0.99	1.00	1.00	1	†	†	§	§	§
6/26	0.93	0.94	0.95	0.95	0.97	0.97	0.98	0.96	0.98	0.98	0.98	0.98	1	§	¶	†	†
6/27	0.91	0.94	0.95	0.95	0.96	0.97	0.98	0.95	0.97	0.97	0.97	0.98	0.98	1	†	†	†
6/28	0.94	0.97	0.97	0.97	0.98	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.98	0.98	1	†	‡
6/29	0.93	0.96	0.96	0.96	0.97	0.97	0.97	0.96	0.98	0.99	0.99	0.99	0.98	0.98	1.00	1	†
6/30	0.94	0.95	0.96	0.95	0.96	0.97	0.97	0.97	0.98	0.98	0.99	0.99	0.98	0.97	0.99	0.99	1
	9/10	9/12	9/19	9/23	9/26	9/29	10/3	10/8	10/10	10/13	10/17	11/7	11/14	11/21	12/5	12/12	12/18
9/10	1	†	§	§	¶	‡	†	†	¶	¶	¶	¶	¶	¶	‡	‡	‡
9/12	0.99	1	§	†	¶	‡	†	†	¶	¶	¶	¶	¶	‡	‡	‡	‡
9/19	0.99	0.99	1	§	¶	¶	§	§	¶	¶	¶	¶	¶	¶	‡	‡	‡
9/23	0.98	0.97	0.98	1	¶	‡	†	†	¶	¶	¶	¶	‡	‡	‡	‡	‡
9/26	0.97	0.98	0.98	0.98	1	†	†	†	‡	‡	‡	‡	†	†	†	†	†
9/29	0.96	0.97	0.98	0.97	0.97	1	†	†	¶	§	¶	§	†	†	†	†	†
10/3	0.92	0.93	0.94	0.92	0.91	0.89	1	†	¶	¶	¶	¶	‡	‡	‡	‡	‡
10/8	0.91	0.9	0.9	0.91	0.92	0.9	¶	¶	¶	¶	¶	¶	‡	‡	‡	‡	‡
10/10	0.9	0.96	0.97	0.91	0.9	0.89	0.97	0.96	1	†	†	†	§	§	§	§	§
10/13	0.85	0.94	0.95	0.87	0.85	0.85	0.92	0.93	0.97	1	†	†	†	†	†	†	†
10/17	0.78	0.92	0.93	0.81	0.78	0.78	0.87	0.88	0.94	0.97	1	†	†	§	†	†	†
11/7	0.93	0.97	0.98	0.93	0.92	0.91	0.98	0.95	0.98	0.94	0.88	1	§	†	†	†	†
11/14	0.92	0.97	0.98	0.93	0.92	0.91	0.98	0.95	0.98	0.94	0.88	1.00	1	†	†	†	†
11/21	0.91	0.95	0.96	0.92	0.91	0.89	0.98	0.97	0.99	0.95	0.91	0.99	0.99	1	†	†	†
12/5	0.93	0.97	0.98	0.93	0.92	0.91	0.98	0.95	0.97	0.93	0.88	0.99	0.99	0.99	1	†	†
12/12	0.92	0.97	0.98	0.93	0.92	0.91	0.98	0.96	0.98	0.94	0.88	0.99	1.00	0.99	0.99	1	†
12/18	0.93	0.97	0.98	0.93	0.93	0.92	0.98	0.97	0.98	0.94	0.89	0.99	1.00	0.99	0.99	1.00	1

Continued next page.

Table 5. Continued.

Deepest monitoring depth (≤ 1.1 m)																	
	6/14	6/15	6/16	6/17	6/18	6/19	6/20	6/21	6/22	6/23	6/24	6/25	6/26	6/27	6/28	6/29	6/30
6/14	1	†	†	†	†	†	†	†	†	†	†	†	†	†	†	†	†
6/15	0.99	1	†	†	†	†	†	†	†	†	†	†	†	†	†	†	†
6/16	0.99	1.00	1	§	†	§	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡
6/17	0.99	1.00	1.00	1	†	†	†	†	†	†	†	†	†	†	†	†	†
6/18	0.99	1.00	1.00	1.00	1	†	†	†	†	†	†	†	†	†	†	†	†
6/19	0.99	1.00	1.00	1.00	1.00	1	†	†	†	†	‡	‡	‡	‡	‡	‡	‡
6/20	0.99	0.99	1.00	1.00	1.00	1.00	1	†	†	†	‡	‡	‡	‡	‡	‡	‡
6/21	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1	†	†	†	‡	‡	‡	‡	‡	‡
6/22	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1.00	1	†	†	‡	‡	‡	‡	‡	‡
6/23	0.99	0.99	0.99	0.99	0.99	1.00	1.00	1.00	1.00	1	†	‡	‡	‡	‡	‡	‡
6/24	0.99	0.99	1.00	1.00	0.99	1.00	1.00	1.00	1.00	1.00	1	†	†	†	†	†	†
6/25	0.98	0.99	0.99	0.99	0.99	0.99	1.00	1.00	1.00	1.00	1.00	1	†	†	†	†	†
6/26	0.97	0.98	0.98	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.99	0.99	1	†	†	†	†
6/27	0.98	0.98	0.99	0.99	0.99	0.99	0.99	0.99	0.99	1.00	1.00	1.00	0.99	1	†	†	†
6/28	0.97	0.98	0.98	0.98	0.98	0.99	0.99	0.99	0.99	1.00	1.00	1.00	0.99	1.00	1	†	†
6/29	0.97	0.98	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.99	0.99	1.00	0.99	1.00	1.00	1	†
6/30	0.97	0.97	0.98	0.98	0.98	0.98	0.99	0.99	0.99	0.99	0.99	0.99	0.99	1.00	1.00	1.00	1
9/10	9/10	9/12	9/19	9/23	9/26	9/29	10/3	10/8	10/10	10/13	10/17	11/7	11/14	11/21	12/5	12/12	12/18
9/10	1	†	†	†	†	†	†	†	†	‡	‡	†	‡	‡	‡	‡	†
9/12	0.99	1	‡	†	†	†	†	‡	†	‡	‡	‡	‡	‡	‡	‡	‡
9/19	0.88	0.95	1	†	†	†	†	‡	†	‡	‡	†	†	†	†	†	†
9/23	0.97	0.97	0.96	1	†	†	†	§	†	‡	§	‡	‡	§	‡	‡	†
9/26	0.87	0.93	0.95	0.93	1	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡
9/29	0.91	0.88	0.89	0.87	0.94	1	†	§	†	‡	§	‡	‡	‡	‡	‡	†
10/3	0.88	0.97	0.97	0.97	0.94	0.87	1	†	†	‡	§	‡	†	†	†	†	†
10/8	0.88	0.95	1.00	0.96	0.95	0.89	0.97	1	†	†	†	†	†	†	†	†	†
10/10	0.87	0.96	0.98	0.95	0.95	0.91	0.98	0.98	1	†	†	†	†	†	†	†	†
10/13	0.86	0.95	0.98	0.96	0.94	0.87	0.98	0.98	0.99	1	†	‡	‡	‡	‡	‡	‡
10/17	0.86	0.96	0.98	0.96	0.94	0.89	0.97	0.98	0.99	0.99	1	†	†	†	†	†	†
11/7	0.91	0.96	0.96	0.96	0.92	0.94	0.95	0.96	0.97	0.96	0.96	1	†	†	†	†	†
11/14	0.92	0.96	0.94	0.97	0.90	0.94	0.94	0.94	0.96	0.94	0.94	0.99	1	†	†	†	†
11/21	0.87	0.97	0.98	0.97	0.95	0.91	0.97	0.98	0.99	0.98	0.99	0.98	0.96	1	†	†	†
12/5	0.91	0.96	0.95	0.98	0.92	0.94	0.95	0.95	0.97	0.96	0.96	0.99	0.99	0.97	1	†	†
12/12	0.93	0.94	0.92	0.98	0.90	0.94	0.93	0.92	0.94	0.93	0.93	0.98	0.99	0.95	0.99	1	†
12/18	0.85	0.95	0.98	0.96	0.94	0.89	0.97	0.98	0.99	0.99	0.99	0.97	0.95	0.99	0.97	0.94	1

† Condition 1.
‡ Condition 2.
§ Condition 3.
‡ Condition 4.

four conditions of temporal stability, while the 0.11- to 0.29-m depth (representing mostly A–B horizon interface) was dominated by Condition 1 (no change) in the first week and then Condition 3 (nonuniform change) in the second week. The deepest monitoring depth (≤ 0.91 – 1.19 m, depending on depth to bedrock) had largely Condition 1 (no change) and Condition 2 (uniform change), suggesting more stable soil moisture

at the deepest monitoring depth. During the complex dry-down and wet-up cycle from September to December, a good mix of the four conditions of temporal stability was observed at all depths, with the deepest monitoring depth having more Condition 1 and less Condition 4 in comparison with the upper soil depths (Table 5). In terms of the moisture storage within the 1.1-m solum, Conditions 3 and 4 dominated throughout

Table 6. Matrices of Pearson's correlation coefficient and four conditions of temporal stability of soil moisture storage within 1.1-m solum at the Shale Hills.

	9/23	9/26	9/29	10/3	10/8	10/10	10/13	10/17	11/7	11/14	11/21	12/5	12/12	12/18
9/23	1	‡	†	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡
9/26	0.99	1	§	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡
9/29	0.99	0.98	1	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡	‡
10/3	0.98	0.99	0.97	1	§	§	§	†	†	†	†	†	†	†
10/8	0.98	0.99	0.97	0.99	1	§	§	†	†	†	†	†	†	†
10/10	0.98	0.99	0.97	0.99	1.00	1	§	†	†	†	†	†	†	†
10/13	0.98	0.99	0.97	0.99	1.00	1.00	1	‡	‡	‡	‡	‡	‡	‡
10/17	0.97	0.98	0.96	0.98	0.99	0.99	1.00	1	‡	‡	‡	‡	‡	‡
11/7	0.98	0.99	0.97	0.99	1.00	1.00	1.00	0.99	1	‡	‡	‡	‡	‡
11/14	0.98	0.99	0.97	0.99	1.00	1.00	1.00	0.99	1.00	1	‡	‡	‡	‡
11/21	0.97	0.99	0.96	0.99	1.00	1.00	1.00	0.99	1.00	1.00	1	‡	‡	‡
12/5	0.97	0.98	0.97	0.99	0.99	0.99	0.99	0.99	1.00	1.00	1.00	1	‡	‡
12/12	0.97	0.99	0.97	0.99	0.99	0.99	0.99	0.99	1.00	1.00	0.99	1.00	1	‡
12/18	0.98	0.99	0.97	0.99	1.00	1.00	1.00	0.99	1.00	1.00	1.00	1.00	0.99	1

† Condition 1.
‡ Condition 2.
§ Condition 3.
‡ Condition 4.

the time from September to December (Table 6), indicating some locations or soils dried or wetted more than others and redistribution occurred significantly within the catchment (e.g., through lateral water movement as discussed in the next section).

In the June dry-down period, the daily dry-down rate ranged from 0.025 to 0.049 m³ m⁻³ in the surface to 0.004 to 0.012 m³ m⁻³ at the deepest monitoring depth (≤ 1.1 m), with a general decreasing rate with depth (Table 7). However, there was no obvious difference in the drying rate among the five soil series, although their absolute moisture contents were quite different and their relative wetness ranking remained temporally persistent (Fig. 8). This is consistent with Conditions 1 and 2 that dominated in this dry-down period (except for 0.11–0.29 m depth where Conditions 1 and 3 dominated). During the wet-up period from 13 July to 15 August, the relative wetness ranking among the three soil series distributed in sloping areas (i.e., the Rushtown, Berks, and Weikert series) remained largely unchanged, but the ranking of the other two soil series located on the valley floor (the Ernest and Blairton series) had some crossovers (Fig. 8). The only soil layer that seemed to respond less to the precipitation and evapotranspiration forcing was the soil horizon below a dense fragipan-like layer (Bt and 2C horizons) in the Ernest series, resulting in the absolute soil moisture content being more time stable throughout the monitoring year (e.g., moisture content at 0.91–1.09 m depth remained around 0.4 m³ m⁻³; Fig. 8).

Table 7. Statistics of a dry-down period during daily measurements from June 14 to 30, 2004 in each of the five soil series.

Soil series	Drying days [†]	Dry-down rate (v/v/d)			
	Mean (Min.–Max.)	Mean	SD	Max.	Min.
% 0–0.06 m depth					
Weikert	69.9 (56.3–87.5)	–0.034	0.005	–0.043	–0.027
Berks	71.9 (68.8–75.0)	–0.030	0.000	–0.030	–0.030
Rushtown	60.7 (50.0–75.0)	–0.040	0.005	–0.048	–0.035
Blairton	75.0 (75.0–75.0)	–0.038	-	-	-
Ernest	62.5 (56.3–75.0)	–0.042	0.011	–0.049	–0.025
Overall	68.0	–0.037	0.005	–0.049	–0.025
% 0.11–0.29 m depth					
Weikert	62.9 (50.0–68.8)	–0.010	0.004	–0.021	–0.005
Berks	78.1 (75.0–81.3)	–0.008	0.002	–0.009	–0.006
Rushtown	67.0 (56.3–75.0)	–0.009	0.003	–0.015	–0.006
Blairton	81.3 (81.3–81.3)	–0.007	-	-	-
Ernest	65.6 (56.3–81.3)	–0.014	0.001	–0.015	–0.012
Overall	71.0	–0.009	0.003	–0.021	–0.005
% 0.51–0.69 m depth					
Weikert	-	-	-	-	-
Berks	56.3 (56.3–56.3)	–0.008	0.003	–0.010	–0.006
Rushtown	59.8 (43.8–87.5)	–0.009	0.002	–0.012	–0.008
Blairton	75.0 (75.0–75.0)	–0.008	-	-	-
Ernest	57.8 (50.0–68.8)	–0.007	0.002	–0.010	–0.005
Overall	61.0	–0.008	0.002	–0.012	–0.005
% 0.91–1.09 m or the deepest depth					
Weikert	62.1 (37.5–87.5)	–0.007	0.003	–0.012	–0.004
Berks	50.0 (43.8–56.3)	–0.007	0.000	–0.007	–0.007
Rushtown	53.6 (43.8–62.5)	–0.008	0.002	–0.012	–0.006
Blairton	43.8 (43.8–43.8)	–0.009	-	-	-
Ernest	53.1 (37.5–62.5)	–0.007	0.001	–0.008	–0.006
Overall	52.5	–0.008	0.002	–0.012	–0.004

[†] There were five rainy days during this period of 17 days (i.e., no rain day 70.5%).

The data from the 16 rain gages collected from September to December 2004 indicated that no obvious spatial distribution in the rainfall existed in this catchment, with an overall similarity between the north- and south-facing slopes and among the five soil series and four landforms (Fig. 9). However, on average, the south-facing slope did receive a slightly higher cumulative rainfall for the period measured, with a mean of 114.7 mm and a CV of 17.1%, while the north-facing slope had a mean cumulative rainfall of 106.0 mm and a CV of 34.7%. Based on area, the south-facing slope contained more deeper soils that were largely distributed in the swales and valley floor (the Rushtown, Ernest, and Blairton series) (Fig. 1). A large pocket of deep Rushtown soil also existed in the eastern end of the south-facing slope that functioned like a water reservoir. More swales in the south-facing slope also facilitated down-slope water movement, because these topographic depressions or hollows are known to be sites of active flow and transport (e.g., Sidle et al., 2000; Tsuboyama et al., 2000; Luxmoore et al., 1990). Consequently, the south-facing slope in this catchment was more active hydrologically than the north-facing slope, possibly draining more water at a faster rate to the stream. This observation is supported by an earlier study by Lynch (1976), who used simulated rainfall to irrigate the entire catchment and proved that the south-facing slope drained much faster than the north-facing slope and contributed more to the early raising limb in the stream hydrograph.

Subsurface Preferential Flow Pathways

The Shale Hills Catchment drainage is influenced by a number of factors, including the 0.05-m-thick sponge-like organic surface layer, horizons of different textures and structures, highly permeable subsoils, subsurface preferential flow pathways, steep slopes, topographic depressions, vegetation dynamics, depth to bedrock, and bedrock orientation and fractures. Because of the sponge-like organic layer at the surface and high permeability in the subsoil, very limited saturation overland flow has been observed in this catchment even during periods of snow melt and large storm events (e.g., during Hurricane Francis and Ivan in September 2004). During such periods, surface runoff was observed only at limited locations at the footslope and toeslope areas (see illustration in Fig. 10). It is widely recognized that the forest litter layer plays an important role in protecting the underlying mineral soils from rain-splash runoff and erosion. By intercepting rainfall, the forest floor serves as a temporal water reservoir and allows more time for infiltration (e.g., Walsh and Voigt, 1977; Kosugi et al., 2001).

Despite the very limited occurrence of saturation overland flow in the Shale Hills Catchment, the stream hydrograph showed rapid response (raising limb) to precipitation forcing, generally about 1 d after each major precipitation event (more than ~20 mm cumulative rainfall in the preceding day or two) (Fig. 3, 8, and 9). This suggests rapid movement of water within the catchment into the stream channel through the subsurface. Figure 9 demonstrates two scenarios of the stream response to

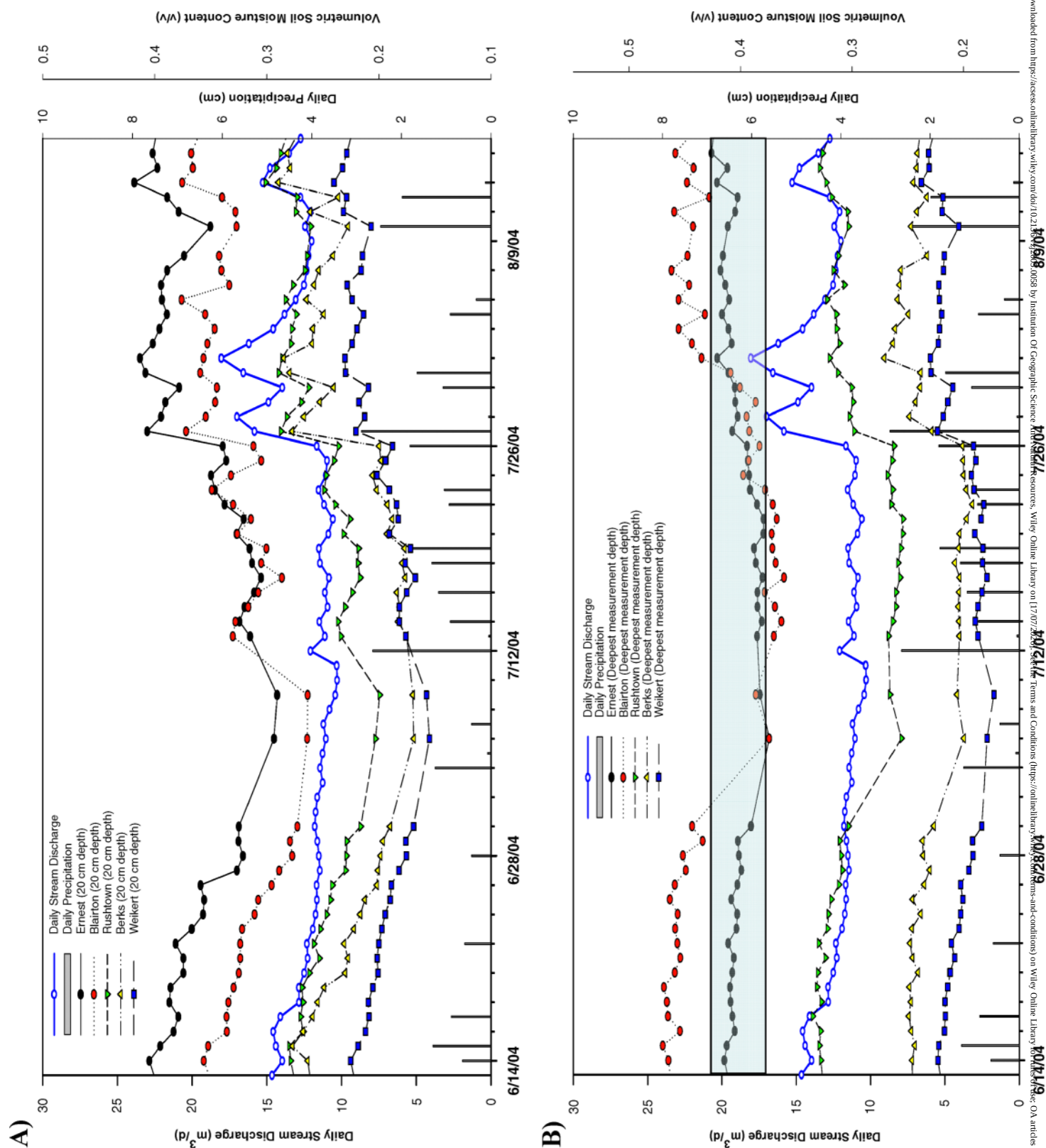


Fig. 8. Averaged daily soil volumetric moisture content at two monitoring depths from 14 June to 15 Aug. 2004 for the five soil series: (A) 0.11 to 0.29 m and (B) the deepest monitoring depth (0.91–1.09 m or shallower depending on the depth to bedrock). Also shown for comparison are daily precipitation and stream discharge at the outlet of the catchment. Note that the lines linking the measurement points are intended for facilitating visual comparison, but do not imply the actual soil moisture change in between the measurement dates.

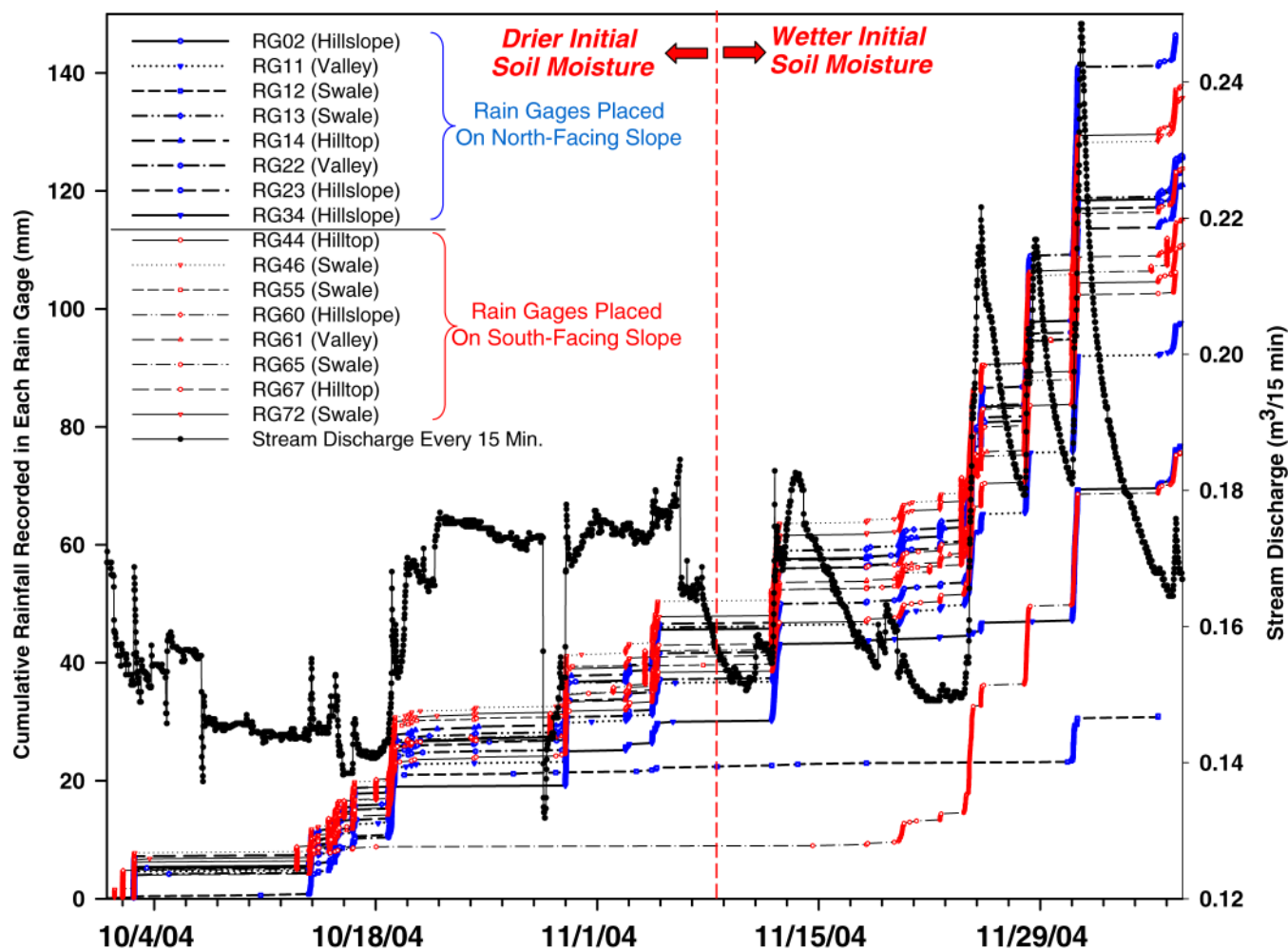


Fig. 9. Comparison of 16 rain gauges' data collected from 29 Sept. to 8 Dec. 2004 between the north- (in blue) and south-facing (in red) slopes. The cumulative rainfall curves showed each tipping (0.2 mm of rain) recorded in the rain gauges. The black dots are stream discharge at 15-min intervals as recorded in stream gauging station at the outlet of the catchment.

precipitation forcing that depended on initial soil moisture status. From early October to early November, the soil moisture was drier, and the stream discharge recorded at 15-min intervals showed an irregular rising and recession limbs (albeit some noises that might exist in the hydrograph), whereas from mid-November to early December, the initial soil moisture was close to saturation, and thus the rising limbs of the stream discharge matched nearly instantaneously with each major rainfall event. During this latter time period, the temporal stability in the subsoil was exclusively Condition 1 (no change) (Table 5), suggesting output at each site equaled input. Lynch (1976) also demonstrated that initial soil moisture was significant in determining the stream hydrograph in the study catchment.

How did the stream hydrograph respond rapidly to precipitation forcing while saturation overland flow was present only very limitedly or not at all? The mechanisms have been debated in the past few decades (e.g., Buttle, 1994; Bonell, 1998; McGlynn et al., 2002). More and more evidence is now pointing to subsurface lateral flow (e.g., Sidle et al., 1995, 2000; Anderson et al., 1997; Uchida et al., 2003). In our study catchment, repeated in

situ observations clearly pointed to the flow pathways of macropore network created by root channels and/or animal burrows, interface between A and B soil horizons, and the soil-bedrock interface (see illustrations in Fig. 10 and 11). These preferential flow pathways were especially evident during the snowmelts and large storm events, which were recorded in real time using videos in this study. The role of subsurface macropore network in hillslope hydrology has been well recognized (e.g., Beven and Germann, 1982; Uchida et al., 1999; Sidle et al., 2000, 2001). In the forested Shale Hills, many chipmunk burrows and tree root channels, especially in the mid to lower portions of the hillslopes, could rapidly transport water downslope during wet periods (e.g., Fig. 10B). Several researchers have also documented the role of soil-bedrock interface as important hillslope flowpath (e.g., Onda et al., 2001; Freer et al., 2002; McGlynn et al., 2002), which was observed in this study (Fig. 10A). Many bubbling seeps were observed at the footslope and toeslope areas, which essentially functioned like pressure-release valves for the water accumulating from upslope areas (Fig. 10C and 11B). One particular seepage point located at the beginning of the perennial

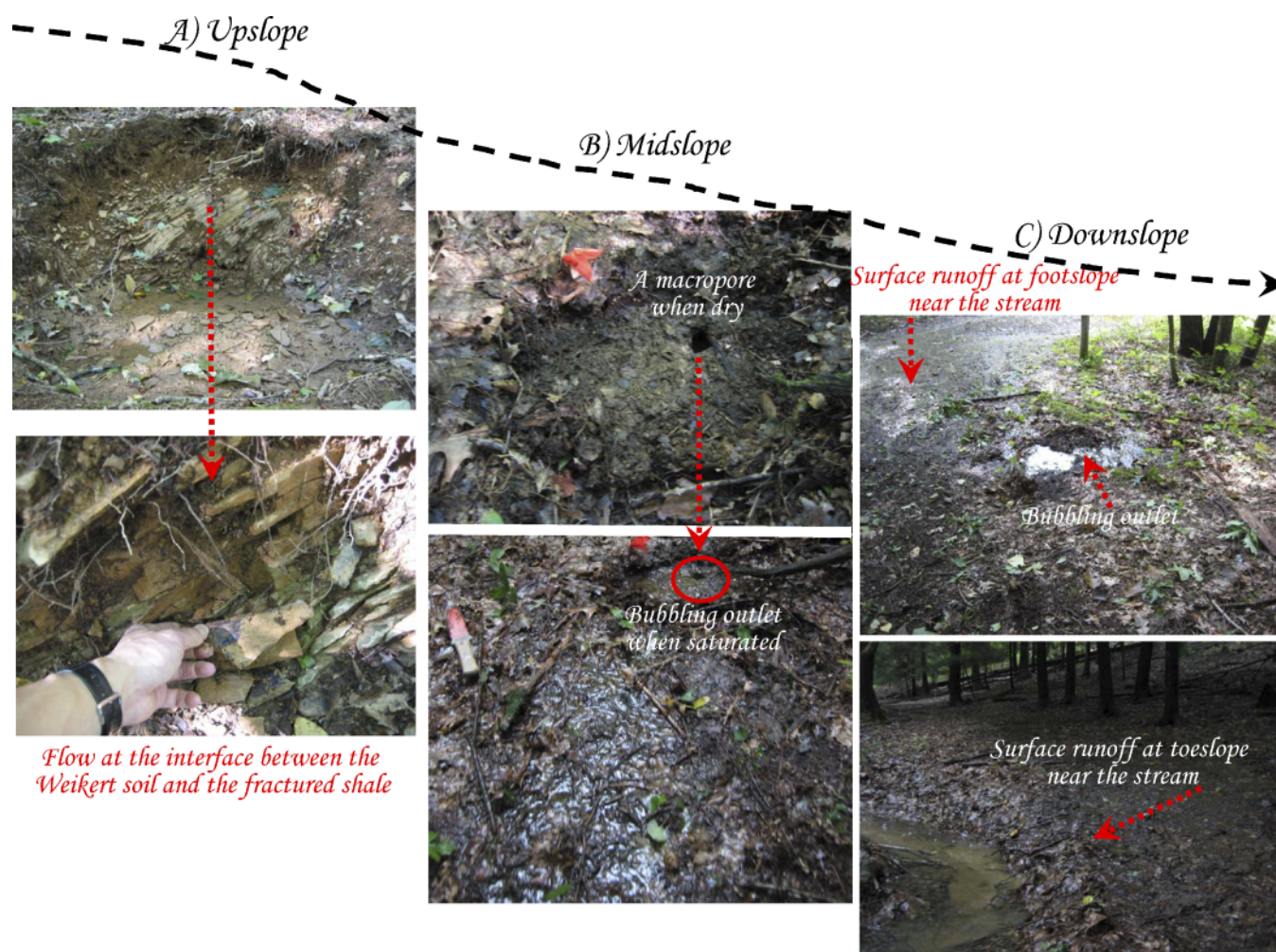


Fig. 10. Illustrations of observed preferential flow pathways at the Shale Hills Catchment along a hillslope of Weikert–Berks–Ernest soil catena: (A) fractured shale in the upslope area (the Weikert soil), (B) a chipmunk burrow about 5 cm in diameter in the midslope area (the Berks soil) when it is dry vs. when it is saturated, and (C) return flow at the footslope and toeslope area (the Ernest soil) showing macropore bubbling seepage and surface runoff near the stream bank.

stream in the catchment (near the sediment fence in Fig. 11B) warrants particular attention because water flows out of this macropore throughout much of the monitoring year. It appears that this particular seepage outlet could have been connected to the water reservoir of the deep Rushtown soil in the eastern end of the catchment and/or another water reservoir in the catchment.

Based on numerous in situ observations, coupled with quantitative soil moisture and hydrologic data collected in this study, we hypothesize that subsurface lateral flow is the main flow mechanism in this catchment. This is a confirmation of a conceptual model of hillslope hydrology suggested by Lin et al. (2006a). The prominence of subsurface lateral flow in this catchment is also supported by the measured soil K_{sat} values in lateral vs. vertical orientations. As shown in Table 1, there were significant differences in lateral and vertical K_{sat} for each of the soil horizons studied, with a ratio of lateral to vertical K_{sat} values ranging from 1.47 to 142.52 in two-thirds (68%) of the soil horizons measured. Most notably was the Blairton series, where we observed significant lateral water movement at the A and B hori-

zon interface particularly during Hurricane Francis and Ivan (Fig. 11A). Such significant lateral movement at the A–B interface in the Blairton soil was also supported by a more than 17-fold decline in vertical K_{sat} values from the A to B horizons (Table 1). In the Shale Hills Catchment, shallow tree rooting systems (branching laterally) and horizontally dipping shale bedrock (11.5–17.1°) favored the development of horizontal preferential flow systems in the soil. Studies in other forested catchments have also shown the lateral branching of tree roots favored planar macropore development in soils (e.g., Noguchi et al., 1997; Sidle et al., 2001). Coupled with the moderate slopes (up to 25–48%), clay-enriched horizons (Bt) in the Blairton and Ernest series, and the shallow soil–bedrock interface throughout much of the catchment hillslopes, subsurface lateral water movement through preferential flow pathways was significant.

CONCLUSIONS

The high spatial variability of soil moisture in the Shale Hills Catchment is largely a result of soil type and land-

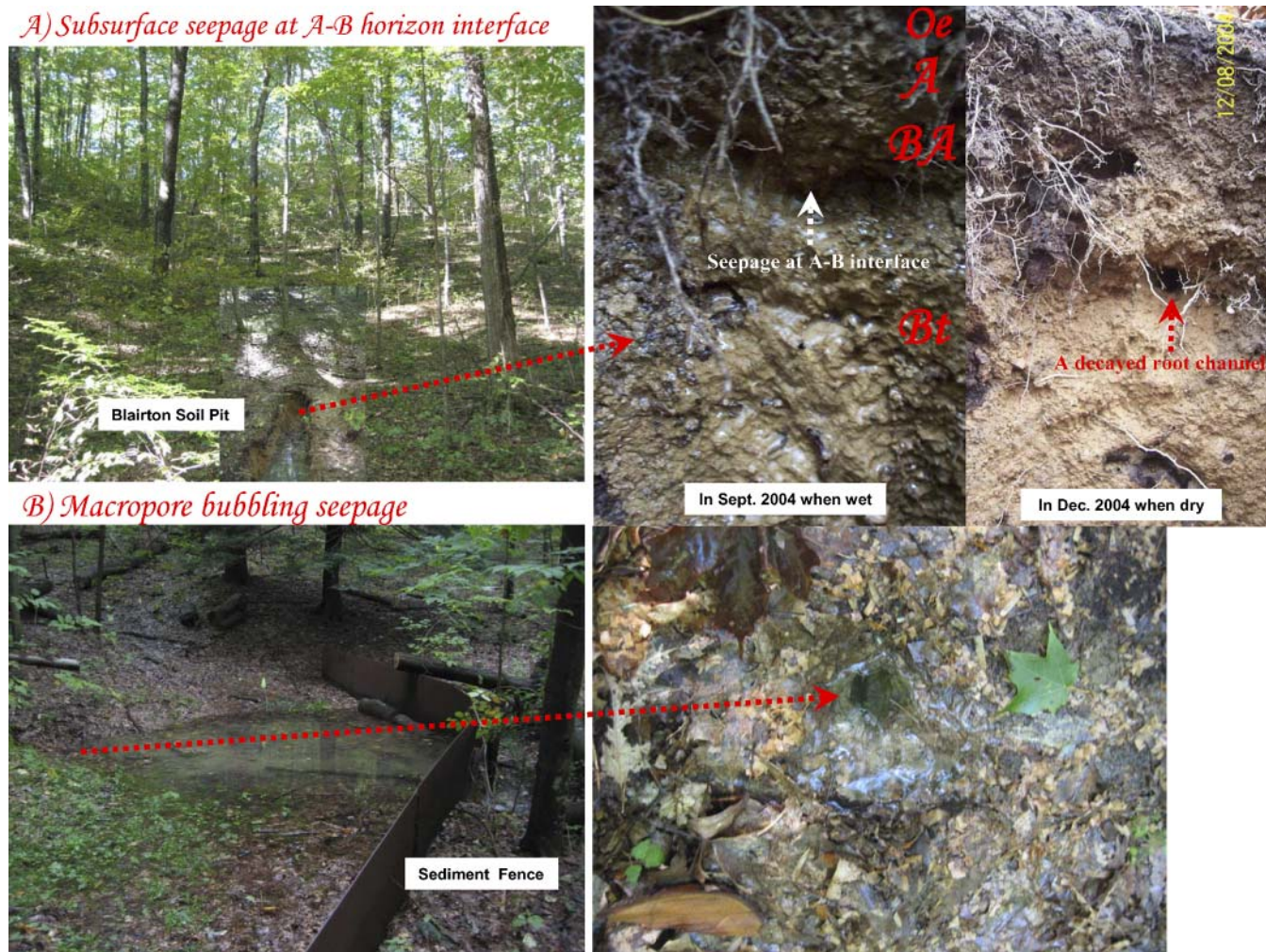


Fig. 11. Illustrations of observed preferential flow pathways at the Shale Hills Catchment: (A) subsurface lateral flow occurred as seepage at the interface between the A and B horizons in the Blairton soil pit during Hurricane Ivan vs. when the soil was dry and (B) macropore bubbling seepage that occurred as uplifting at the beginning of the perennial stream in the catchment (near the sediment fence). These two preferential flow pathways were particularly prominent during storm events (e.g., Hurricane Francis and Ivan in September 2004). See Fig. 2 regarding the locations of these two sites within the catchment.

form controls rather than a random process. This has led to a considerable temporal stability of soil moisture spatial pattern in the catchment, including the ranking stability of the 77 monitoring sites throughout the monitoring year and the four conditions of temporal stability that were related to soil moisture and hydrologic dynamics. The five soil series, while having distinctly different moisture storage capacities, exhibited temporal persistence in the spatial distribution and depth function of their relative wetness ranking. Among the four landform units in this catchment, the valley floor and swales consistently stored more soil moisture throughout the year.

This study demonstrates the synergies of integrating pedologic and hydrologic knowledge in enhancing the understanding of soil moisture and hydrologic dynamics in a humid forested catchment. The results highlight the importance of the soil zone in hillslope/catchment hydrology, including the distribution of different soil types, soil thickness, soil layering, soil pore network, and preferential flow pathways. Subsurface lateral flow was prominent as a result of an environment favorable for

lateral soil pore development, coupled with moderate slopes, hydrologically active swales, clay-enriched subsoil horizons, and shallow soil–bedrock interface. This explains the rapid response in the stream hydrograph to precipitation forcing while saturation overland flow was very limited in this catchment. Because of more extensively distributed deeper soils and swales, coupled with subsurface lateral flow pathways and slightly higher cumulative rainfall, the south-facing slope was hydrologically more active than the north-facing slope in terms of draining more water at a faster rate to the stream. The hydrogeologic approach illustrated in this study is beneficial in enhancing the mapping, monitoring, and modeling of landscape hydrology at the hillslope and catchment scales.

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